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Compositions and methods for detecting or quantifying human parainfluenza virus 2

Abstract

Compositions, methods, kits, and uses are provided for detecting or quantifying an Human Parainfluenza virus 1 (HPIV-1), HPIV-2, HPIV-3, and/or HPIV-4 nucleic acid, e.g., using nucleic acid amplification and hybridization assays. In some embodiments, the compositions, methods, kits, and uses target the HN gene of HPIV-1, HPIV-2, and/or HPIV-3 and/or the NP gene of HPIV-4.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 17/072,890, filed Oct. 16, 2020, which is a divisional of U.S. application Ser. No. 15/934,273, filed Mar. 23, 2018, now issued as U.S. Pat. No. 10,844,425, which claims the benefit of U.S. Provisional Patent Application No. 62/476,435, filed Mar. 24, 2017. Each of the foregoing applications is incorporated herein by reference for all purposes.

REFERENCE TO SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted in XML format via Patent Center and is hereby incorporated by reference in its entirety. Said XML Copy, created on Jun. 13, 2024, is named 4340-P28US_DIV3_SL and is 182,419 bytes in size.

TECHNICAL FIELD

(2) The embodiments herein are directed to the field of detecting infectious agents, more specifically by using compositions and methods to detect viruses including Human Parainfluenza Viruses 1, 2, 3, and 4 (HPIV-1, HPIV-2, HPIV-3, and HPIV-4).

SUMMARY

(3) Human Parainfluenza Viruses are members of the Paramyxoviridae family and are negative-sense, single-stranded RNA viruses. HPIV types 1-4 have been identified, with type 4 containing subtypes 4a and 4b. HPIVs can cause cold-like symptoms, respiratory tract illnesses, croup (especially HPIV-1 and 2), and bronchiolitis, bronchitis, and/or pneumonia (especially HPIV-3). HPIVs are spread in modes common to respiratory viruses, such as through the air including by coughing or sneezing, through close personal contact including handshakes or other touching, or through touching objects or surfaces exposed to the virus.

(4) There are no known vaccines or specific antiviral treatment for HPIV infections, but symptoms can be managed. Given the variety of symptoms that can be caused by HPIV and their similarity to other respiratory infections such as colds, influenza, and bacterial pneumonia, it is important to rapidly identify HPIV infections as such to facilitate appropriate treatments and avoid administering unnecessary or ineffective medications.

(5) Nucleic acid-based detection techniques, including amplification-based approaches, e.g., TaqMan™ RT-PCR, provide rapid detection and quantification of HPIVs. However, the heterogeneity of HPIV sequences across and within the various types and subtypes can cause difficulties with the sensitivity and/or specificity of such techniques. Accordingly, there is a need for compositions and methods that allow sensitive and specific detection and quantification of HPIVs. This disclosure aims to meet these needs, provide other benefits, or at least provide the public with a useful choice.

(6) Accordingly, provided herein is a composition or kit comprising at least first and second amplification oligomers,

wherein: the first amplification oligomer and second amplification oligomer are configured to amplify an HPIV-3 amplicon of at least about 50 nucleotides in length comprising at least one Human Parainfluenza Virus 3 (HPIV-3) position located in the range of positions 1295-1305, 1350-1360, and/or 1380-1390.

(7) Also provided herein is a composition or kit comprising at least first and second amplification oligomers, wherein: (a) the first amplification oligomer is configured to hybridize to a site comprising HPIV-3 position 1270 and the second amplification oligomer is configured to hybridize to a site comprising HPIV-3 position 1355, and the first and second amplification oligomers are configured to produce an HPIV-3 amplicon; or (b) the first amplification oligomer is configured to hybridize to a site comprising HPIV-3 position 1355 and the second amplification oligomer is configured to hybridize to a site comprising HPIV-3 position 1437, and the first and second amplification oligomers are configured to produce an HPIV-3 amplicon.

(8) Also provided herein is a method of detecting HPIV-3 in a sample, comprising: contacting the sample with at least first and second amplification oligomers, thereby forming a composition, performing a nucleic acid amplification reaction in the composition which produces an HPIV-3 amplicon of at least about 50 nucleotides in length in the presence of an HPIV-3 nucleic acid, and detecting the presence or absence of the at least one amplicon, wherein: the first amplification oligomer and second amplification oligomer are configured to amplify an HPIV-3 amplicon comprising at least one Human Parainfluenza Virus 3 (HPIV-3) position located within positions 1295-1305, 1350-1360, and/or 1380-1390.

(9) Also provided herein is a method of detecting HPIV-3 in a sample, comprising: contacting the sample with at least first and second amplification oligomers, thereby forming a composition, performing a nucleic acid amplification reaction in the composition which produces an HPIV-3 amplicon in the presence of an HPIV-3 nucleic acid, and detecting the presence or absence of the at least one amplicon, wherein: (a) the first amplification oligomer is configured to hybridize to a site comprising HPIV-3 position 1270 and the second amplification oligomer is configured to hybridize to a site comprising HPIV-3 position 1355, and the first and second amplification oligomers are configured to produce an HPIV-3 amplicon; or (b) the first amplification oligomer is configured to hybridize to a site comprising HPIV-3 position 1355 and the second amplification oligomer is configured to hybridize to a site comprising HPIV-3 position 1437, and the first and second amplification oligomers are configured to produce an HPIV-3 amplicon.

(10) In some embodiments, the amplicon comprises HPIV-3 position 1305. In some embodiments, the amplicon comprises HPIV-3 position 1355. In some embodiments, the amplicon comprises HPIV-3 position 1380.

(11) In some embodiments, the first oligomer competes for hybridization to an HPIV-3 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 25. In some embodiments, the first oligomer comprises the sequence of SEQ ID NO: 25 with up to two mismatches. In some embodiments, the first oligomer comprises the sequence of SEQ ID NO: 25.

(12) In some embodiments, the second oligomer competes for hybridization to an HPIV-3 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 36. In some embodiments, wherein the second oligomer comprises the sequence of SEQ ID NO: 36 with up to two mismatches. In some embodiments, wherein the second oligomer comprises the sequence of SEQ ID NO: 36.

(13) In some embodiments, wherein the first oligomer competes for hybridization to an HPIV-3 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 28. In some embodiments, wherein the first oligomer comprises the sequence of SEQ ID NO: 28 with up to two mismatches. In some embodiments, wherein the first oligomer comprises the sequence of SEQ ID NO: 28.

(14) In some embodiments, wherein the second oligomer competes for hybridization to an HPIV-3 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 37. In some embodiments, wherein the second oligomer comprises the sequence of SEQ ID NO: 37 with up to two mismatches. In some embodiments, wherein the second oligomer comprises the sequence of SEQ ID NO: 37.

(15) In some embodiments, the first oligomer comprises at least one 5-methylcytosine residue. In some embodiments, at least about half of the C residues in the first oligomer are 5-methylated. In some embodiments, substantially all of the C residues in the first oligomer are 5-methylated. In some embodiments, the second oligomer comprises at least one 5-methylcytosine. In some embodiments, at least about half of the C residues in the second oligomer are 5-methylated. In some embodiments, substantially all of the C residues in the second oligomer are 5-methylated.

(16) In some embodiments, a composition or kit disclosed herein further comprises a third oligomer configured to hybridize to the amplicon, and the third oligomer comprises a detectable label. In some embodiments of a method disclosed herein, the presence or absence of the amplicon is detected according to the occurrence or non-occurrence of hybridization of the third oligomer to the amplicon.

(17) In some embodiments of a method disclosed herein, an HPIV-3 nucleic acid is quantified by monitoring production of the HPIV-3 amplicon.

(18) In some embodiments, the third oligomer competes for hybridization to an HPIV-3 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 31, 32, 33, 34, or 35. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 31 with up to two mismatches. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 31. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 32 with up to two mismatches. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 32. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 33 with up to two mismatches. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 33. In some embodiments, the third

oligomer comprises the sequence of SEQ ID NO: 34 with up to two mismatches. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 34. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 35 with up to two mismatches. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 35.

(19) In some embodiments, the third oligomer competes for hybridization to an HPIV-3 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 26, 27, 28, 29, or 30. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 26 with up to two mismatches. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 26. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 27 with up to two mismatches. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 27. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 28 with up to two mismatches. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 28. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 29 with up to two mismatches. In some embodiments, the third oligomer comprises the sequence of SEQ ID NO: 29.

(20) In some embodiments, the third oligomer comprises at least one 5-methylcytosine. In some embodiments, at least about half of the C residues in the third oligomer are 5-methylated. In some embodiments, substantially all of the C residues in the third oligomer are 5-methylated.

(21) In some embodiments, the composition or kit further comprises a fourth oligomer different from the third oligomer, and the fourth oligomer is configured to hybridize to the amplicon and comprises a detectable label. In some embodiments, the fourth oligomer comprises the sequence of SEQ ID NO: 31, 32, 33, 34, or 35. In some embodiments, the fourth oligomer comprises the sequence of SEQ ID NO: 26, 27, 28, 29, or 30. In some embodiments, the fourth oligomer comprises at least one 5-methylcytosine. In some embodiments, at least about half of the C residues in the fourth oligomer are 5-methylated. In some embodiments, substantially all of the C residues in the fourth oligomer are 5-methylated.

(22) In some embodiments, a composition or kit disclosed herein further comprises a first Human Parainfluenza Virus 1 (HPIV-1) amplification oligomer and a second HPIV-1 amplification oligomer, wherein: the first HPIV-1 amplification oligomer and second HPIV-1 amplification oligomer are configured to amplify an HPIV-1 amplicon of at least about 50 nucleotides in length comprising at least one HPIV-1 position located within HPIV-1 positions 330-490 or 960-1100.

(23) Also provided herein is a composition or kit comprising at least a first HPIV-1 amplification oligomer and a second HPIV-1 amplification oligomer, wherein: the first HPIV-1 amplification oligomer and second HPIV-1 amplification oligomer are configured to amplify an HPIV-1 amplicon of at least about 50 nucleotides in length comprising at least one HPIV-1 position located within HPIV-1 positions 330-490 or 960-1100.

(24) Also provided herein is a method of detecting HPIV-1 in a sample, comprising: contacting the sample with at least first and second amplification oligomers, thereby forming a composition, performing a nucleic acid amplification reaction in the composition which produces an HPIV-1 amplicon in the presence of an HPIV-1 nucleic acid, and detecting the presence or absence of the at least one amplicon, wherein: the first HPIV-1 amplification oligomer and second HPIV-1 amplification oligomer are configured to amplify an HPIV-1 amplicon of at least about 50 nucleotides in length comprising at least one HPIV-1 position located within HPIV-1 positions 330-490 or 960-1100.

(25) In some embodiments, the first HPIV-1 amplification oligomer is configured to hybridize to a site comprising HPIV-1 position 970 and the second HPIV-1 amplification oligomer is configured to hybridize to a site comprising HPIV-1 position 1098.

(26) In some embodiments, the first HPIV-1 amplification oligomer competes for hybridization to an HPIV-1 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 6. In some embodiments, the first HPIV-1 oligomer comprises the sequence of SEQ ID NO: 6 with up to two mismatches. In some embodiments, the first HPIV-1 oligomer comprises the sequence of SEQ ID NO: 6.

(27) In some embodiments, the second HPIV-1 amplification oligomer competes for hybridization to an HPIV-1 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 24. In some embodiments, the second HPIV-1 oligomer comprises the sequence of SEQ ID NO: 24 with up to two mismatches. In some embodiments, the second HPIV-1 oligomer comprises the sequence of SEQ ID NO: 24.

(28) In some embodiments, the first HPIV-1 amplification oligomer is configured to hybridize to a site comprising HPIV-1 position 331 and the second HPIV-1 amplification oligomer is configured to hybridize to a site comprising HPIV-1 position 486.

(29) In some embodiments, the first HPIV-1 amplification oligomer competes for hybridization to an HPIV-1 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 17. In some embodiments, the first HPIV-1 oligomer comprises the sequence of SEQ ID NO: 17 with up to two mismatches. In some embodiments, the first HPIV-1 oligomer comprises the sequence of SEQ ID NO: 17.

(30) In some embodiments, the second HPIV-1 amplification oligomer competes for hybridization to an HPIV-1 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 22. In some embodiments, the second HPIV-1 oligomer comprises the sequence of SEQ ID NO: 22 with up to two mismatches. In some embodiments, the second HPIV-1 oligomer comprises the sequence of SEQ ID NO: 22.

(31) In some embodiments, the composition or kit further comprises a third HPIV-1 oligomer configured to hybridize to the HPIV-1 amplicon, and the third oligomer comprises a detectable label.

(32) In some embodiments, the third HPIV-1 oligomer competes for hybridization to an HPIV-1 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 5, 7, 8, 9, 12, 13, 14, 15, 16, 18, 19, 20, 21, 23, or 125. In some embodiments, the 75 or 76, wherein the third HPIV-1 oligomer comprises the sequence of SEQ ID NO: 5, 7, 8, 9, 12, 13, 14, 15, 16, 18, 19, 20, 21, 23, or 125 with up to two mismatches. In some embodiments, the third HPIV-1 oligomer comprises the sequence of SEQ ID NO: 5, 7, 21, 23, or 125. In some embodiments, the third HPIV-1 oligomer comprises the sequence of SEQ ID NO: 8, 9, 20, or 125. In some embodiments, the third HPIV-1 oligomer comprises the sequence of SEQ ID NO: 12, 13, 14, or 15. In some embodiments, the third HPIV-1 oligomer comprises the sequence of SEQ ID NO: 16, 18, or 19.

(33) In some embodiments, the third HPIV-1 oligomer competes for hybridization to an HPIV-1 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 10. In some embodiments, the third HPIV-1 oligomer comprises the sequence of SEQ ID NO: 10 with up to two mismatches. In some embodiments, the third HPIV-1 oligomer comprises the sequence of SEQ ID NO: 10.

(34) In some embodiments, the third HPIV-1 oligomer competes for hybridization to an HPIV-1 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 11. In some embodiments, the third HPIV-1 oligomer comprises the sequence of SEQ ID NO: 11 with up to two mismatches. In some embodiments, the third HPIV-1 oligomer comprises the sequence of SEQ ID NO: 11.

(35) In some embodiments, at least one, at least two, or each of the first, second, and third HPIV-1 oligomers comprise at least one 5-methylcytosine. In some embodiments, at least about half of the C residues in the first, second, and/or third HPIV-1 oligomers are 5-methylated. In some embodiments, substantially all of the C residues in the first, second, and/or third HPIV-1 oligomers are 5-methylated.

(36) In some embodiments, a kit or composition further comprises a first Human Parainfluenza Virus 1 (HPIV-2) amplification oligomer and a second HPIV-2 amplification oligomer, wherein: the first HPIV-2 amplification oligomer and second HPIV-2 amplification oligomer are configured to amplify an HPIV-2 amplicon of at least about 50 nucleotides in length comprising at least one HPIV-2 position located within HPIV-2 positions 1600-1700.

(37) Also provided herein is a composition or kit comprising at least a first HPIV-2 amplification oligomer and a second HPIV-2 amplification oligomer, wherein: the first HPIV-2 amplification oligomer and second HPIV-2 amplification oligomer are configured to amplify an HPIV-2 amplicon of at least about 50 nucleotides in length comprising at least one HPIV-2 position located within HPIV-2 positions 1600-1700.

(38) Also provided herein is a method of detecting HPIV-2 in a sample, comprising: contacting the sample with at least first and second amplification oligomers, thereby forming a composition, performing a nucleic acid amplification reaction in the composition which produces an HPIV-2 amplicon in the presence of an HPIV-2 nucleic acid, and detecting the presence or absence of the at least one amplicon, wherein: the first HPIV-2 amplification oligomer and second HPIV-2 amplification oligomer are configured to amplify an HPIV-2 amplicon of at least about 50 nucleotides in length comprising at least one HPIV-2 position located within HPIV-2 positions 1600-1700.

(39) In some embodiments, the first HPIV-2 amplification oligomer is configured to hybridize to a site comprising HPIV-2 position 1603 and the second HPIV-2 amplification oligomer is configured to hybridize to a site comprising HPIV-2 position 1698.

(40) In some embodiments, the first HPIV-2 amplification oligomer competes for hybridization to an HPIV-2 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 44. In some embodiments, the first HPIV-2 oligomer comprises the sequence of SEQ ID NO: 44 with up to two mismatches. In some embodiments, the first HPIV-2 oligomer comprises the sequence of SEQ ID NO: 44.

(41) In some embodiments, the second HPIV-2 amplification oligomer competes for hybridization to an HPIV-2 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 40. In some embodiments, the second HPIV-2 oligomer comprises the sequence of SEQ ID NO: 40 with up to two mismatches. In some embodiments, the second HPIV-2 oligomer comprises the sequence of SEQ ID NO: 40.

(42) In some embodiments, the composition or kit further comprises a third HPIV-2 oligomer configured to hybridize to the HPIV-2 amplicon, and the third oligomer comprises a detectable label.

(43) In some embodiments, the third HPIV-2 oligomer competes for hybridization to an HPIV-2 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 38, 39, 41, 42, 43, or 45. In some embodiments, the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 38, 39, 41, 42, 43, or 45 with up to two mismatches. In some embodiments, the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 45. In some embodiments, the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 43. In some embodiments, the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 42. In some embodiments, the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 41. In some embodiments, the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 39. In some embodiments, the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 38.

(44) In some embodiments, at least one, at least two, or each of the first, second, and third HPIV-2 oligomers comprise at least one 5-methylcytosine. In some embodiments, at least about half of the C residues in the first, second, and/or third HPIV-2 oligomers are 5-methylated. In some embodiments, substantially all of the C residues in the first, second, and/or third HPIV-2 oligomers are 5-methylated.

(45) In some embodiments, a composition or kit disclosed herein further comprises a first Human Parainfluenza Virus 1 (HPIV-4) amplification oligomer and a second HPIV-4 amplification oligomer, wherein: the first HPIV-4 amplification

oligomer and second HPIV-4 amplification oligomer are configured to amplify an HPIV-4 amplicon of at least about 50 nucleotides in length comprising at least one HPIV-4 position located within HPIV-4 positions 620-740, 2130-2410, 2520-3040, or 10090-11980.

(46) Also provided herein is a composition or kit comprising at least a first HPIV-4 amplification oligomer and a second HPIV-4 amplification oligomer, wherein: the first HPIV-4 amplification oligomer and second HPIV-4 amplification oligomer are configured to amplify an HPIV-4 amplicon of at least about 50 nucleotides in length comprising at least one HPIV-4 position located within HPIV-4 positions 620-740, 2130-2410, 2520-3040, or 10090-11980.

(47) Also provided herein is a method of detecting HPIV-4 in a sample, comprising: contacting the sample with at least first and second amplification oligomers, thereby forming a composition, performing a nucleic acid amplification reaction in the composition which produces an HPIV-4 amplicon in the presence of an HPIV-4 nucleic acid, and detecting the presence or absence of the at least one amplicon, wherein: the first HPIV-4 amplification oligomer and second HPIV-4 amplification oligomer are configured to amplify an HPIV-4 amplicon of at least about 50 nucleotides in length comprising at least one HPIV-4 position located within HPIV-4 positions 620-740, 2130-2410, 2520-3040, or 10090-11980.

(48) In some embodiments, the first HPIV-4 amplification oligomer is configured to hybridize to a site comprising HPIV-4 position 10094, 10205, or 10248, and the second HPIV-4 amplification oligomer is configured to hybridize to a site comprising HPIV-4 position 11685, 11739, or 11981.

(49) In some embodiments, the first HPIV-4 amplification oligomer competes for hybridization to an HPIV-4 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 79, 80, 81, 82, 97, 98, 99, or 100. In some embodiments, the first HPIV-4 oligomer comprises the sequence of SEQ ID NO: 79, 80, 81, 82, 97, 98, 99, or 100 with up to two mismatches. In some embodiments, the first HPIV-4 oligomer comprises the sequence of SEQ ID NO: 79, 80, 81, 82, 97, 98, 99, or 100.

(50) In some embodiments, the second HPIV-4 amplification oligomer competes for hybridization to an HPIV-4 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 54, 63, 64, 113, 114, or 115. In some embodiments, the second HPIV-4 oligomer comprises the sequence of SEQ ID NO: 54, 63, 64, 113, 114, or 115 with up to two mismatches. In some embodiments, the second HPIV-4 oligomer comprises the sequence of SEQ ID NO: 54, 63, 64, 113, 114, or 115.

(51) In some embodiments, the composition or kit further comprises a third HPIV-4 oligomer configured to hybridize to the HPIV-4 amplicon, and the third oligomer comprises a detectable label.

(52) In some embodiments, the third HPIV-4 oligomer competes for hybridization to an HPIV-4 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 53, 55, 58, 60, 62, 65, 67, 68, 76, 83, 84, 85, 86, 92, 94, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, or 112. In some embodiments, the third HPIV-4 oligomer comprises the sequence of SEQ ID NO: 53, 55, 58, 60, 62, 65, 67, 68, 76, 83, 84, 85, 86, 92, 94, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, or 112 with up to two mismatches. In some embodiments, the third HPIV-4 oligomer comprises the sequence of SEQ ID NO: 53, 55, 58, 60, 62, 65, 67, 68, 76, 83, 84, 85, 86, 92, 94, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, or 112.

(53) In some embodiments, the first HPIV-4 amplification oligomer is configured to hybridize to a site comprising HPIV-4 position 625 and the second HPIV-4 amplification oligomer is configured to hybridize to a site comprising HPIV-4 position 738.

(54) In some embodiments, the first HPIV-4 amplification oligomer competes for hybridization to an HPIV-4 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 78. In some embodiments, the first HPIV-4 oligomer comprises the sequence of SEQ ID NO: 78 with up to two mismatches. In some embodiments, the first HPIV-4 oligomer comprises the sequence of SEQ ID NO: 78.

(55) In some embodiments, the second HPIV-4 amplification oligomer competes for hybridization to an HPIV-4 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 56. In some embodiments, the second HPIV-4 oligomer comprises the sequence of SEQ ID NO: 56 with up to two mismatches. In some embodiments, the second HPIV-4 oligomer comprises the sequence of SEQ ID NO: 56.

(56) In some embodiments, the composition or kit further comprises a third HPIV-4 oligomer configured to hybridize to the HPIV-4 amplicon, and the third oligomer comprises a detectable label.

(57) In some embodiments, the third HPIV-4 oligomer competes for hybridization to an HPIV-4 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 71. In some embodiments, the third HPIV-4 oligomer comprises the sequence of SEQ ID NO: 71 with up to two mismatches. In some embodiments, the third HPIV-4 oligomer comprises the sequence of SEQ ID NO: 71.

(58) In some embodiments, the first HPIV-4 amplification oligomer is configured to hybridize to a site comprising HPIV-4 position 2128, 2352, 2386, 2388, 2404, or 2524 and the second HPIV-4 amplification oligomer is configured to hybridize to a site comprising HPIV-4 position 2907, 3006, or 3039. In some embodiments, the first HPIV-4 amplification oligomer competes for hybridization to an HPIV-4 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 72 or 124. In some embodiments, the first HPIV-4 oligomer comprises the sequence of SEQ ID NO: 72 or 124 with up to two mismatches. In some embodiments, the first HPIV-4 oligomer comprises the sequence of SEQ ID NO: 72 or 124.

(59) In some embodiments, the second HPIV-4 amplification oligomer competes for hybridization to an HPIV-4 nucleic

acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 50, 51, 52, 93, 95, or 96. In some embodiments, the second HPIV-4 oligomer comprises the sequence of SEQ ID NO: 50, 51, 52, 93, 95, or 96 with up to two mismatches. In some embodiments, the second HPIV-4 oligomer comprises the sequence of SEQ ID NO: 50, 51, 52, 93, 95, or 96.

(60) In some embodiments, the composition or kit further comprises a third HPIV-4 oligomer configured to hybridize to the HPIV-4 amplicon, and the third oligomer comprises a detectable label.

(61) In some embodiments, the third HPIV-4 oligomer competes for hybridization to an HPIV-4 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 46, 47, 48, 49, 59, 61, 69, 70, 72, 73, 74, 75, 77, 88, 90, 91, 116, 117, 118, 119, 121, 122, or 123. In some embodiments, the third HPIV-4 oligomer comprises the sequence of SEQ ID NO: 46, 47, 48, 49, 59, 61, 69, 70, 72, 73, 74, 75, 77, 88, 90, 91, 116, 117, 118, 119, 121, 122, or 123 with up to two mismatches. In some embodiments, the third HPIV-4 oligomer comprises the sequence of SEQ ID NO: 46, 47, 48, 49, 59, 61, 69, 70, 72, 73, 74, 75, 77, 88, 90, 91, 116, 117, 118, 119, 121, 122, or 123.

(62) In some embodiments, at least one, at least two, or each of the first, second, and third HPIV-4 oligomers comprise at least one 5-methylcytosine. In some embodiments, at least about half of the C residues in the first, second, and/or third HPIV-4 oligomers are 5-methylated. In some embodiments, substantially all of the C residues in the first, second, and/or third HPIV-4 oligomers are 5-methylated.

(63) In some embodiments of a method disclosed herein, the nucleic acid amplification reaction comprises thermal cycling. In some embodiments of a method disclosed herein, the nucleic acid amplification reaction comprises reverse transcription. In some embodiments of a method disclosed herein, the nucleic acid amplification reaction comprises RT-PCR. In some embodiments of a method disclosed herein, at least one detectable label is fluorescent. In some embodiments, at least one oligomer comprising a fluorescent label further comprises a quencher. In some embodiments, at least one oligomer comprising a fluorescent label is a non-extendable oligomer. In some embodiments of a method disclosed herein, the nucleic acid amplification reaction comprises PCR with a polymerase with 5'-to-3' exonuclease activity, and detecting an amplicon using a probe oligomer comprising a fluorophore and a quencher, wherein exonucleolysis of the probe by the polymerase reduces quenching of fluorescence by the quencher.

(64) In some embodiments, at least one, at least two, or at least three oligomers comprise at least one non-Watson Crick (NWC) position. In some embodiments, at least one, two, three, or four of oligomers configured to hybridize to an HPIV-1, HPIV-2, HPIV-3, or HPIV-4 each comprise at least one NWC position.

(65) In some embodiments, at least one, at least two, or at least three oligomers comprise a degenerate position. In some embodiments, at least one, two, three, or four of oligomers configured to hybridize to an HPIV-1, HPIV-2, HPIV-3, or HPIV-4 each comprise at least one degenerate position.

(66) In some embodiments, the first and second amplification oligomers, the first and second HPIV-1 amplification oligomers, the first and second HPIV-2 amplification oligomers, and/or the first and second HPIV-4 amplification oligomers are comprise target-hybridizing regions from about 10-60 bases in length, about 14-50 bases in length, about 14-40 bases in length, about 14-35 bases in length, or about 15-30 bases in length.

(67) In some embodiments, the HPIV-3 amplicon, the HPIV-1 amplicon, the HPIV-2 amplicon, and/or the HPIV-4 amplicon are about 50-1000 nucleotides, about 50-100 nucleotides, about 100-200 nucleotides, about 200-300 nucleotides, about 300-400 nucleotides, about 400-500 nucleotides, about 500-600 nucleotides, about 600-700 nucleotides, about 700-800 nucleotides, about 800-900 nucleotides, about 900-1000 nucleotides, or about 1000-2000 nucleotides in length.

(68) In some embodiments, a composition disclosed herein is aqueous, frozen, or lyophilized.

(69) Also disclosed herein is a use of a composition or kit disclosed herein for detecting or quantifying an HPIV-1, HPIV-2, HPIV-3, and/or HPIV-4 nucleic acid in a sample.

Description

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

Definitions

(1) Before describing the present teachings in detail, it is to be understood that the disclosure is not limited to specific compositions or process steps, as such may vary. It should be noted that, as used in this specification and the appended claims, the singular form “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Thus, for example, reference to “an oligomer” includes a plurality of oligomers and the like. The conjunction “or” is to be interpreted in the inclusive sense, i.e., as equivalent to “and/or,” unless the inclusive sense would be unreasonable in the context.

(2) It will be appreciated that there is an implied “about” prior to the temperatures, concentrations, times, etc. discussed in the present disclosure, such that slight and insubstantial deviations are within the scope of the present teachings herein. In general, the term “about” indicates insubstantial variation in a quantity of a component of a composition not having any significant effect on the activity or stability of the composition. All ranges are to be interpreted as encompassing the endpoints in the absence of express exclusions such as “not including the endpoints”; thus, for example, “within 10-15” includes the values 10 and 15. Also, the use of “comprise,” “comprises,” “comprising,” “contain,” “contains,” “containing,” “include,” “includes,” and “including” are not intended to be limiting. It is to be understood that both the

foregoing general description and detailed description are exemplary and explanatory only and are not restrictive of the teachings. To the extent that any material incorporated by reference is inconsistent with the express content of this disclosure, the express content controls.

(3) Unless specifically noted, embodiments in the specification that recite “comprising” various components are also contemplated as “consisting of” or “consisting essentially of” the recited components; embodiments in the specification that recite “consisting of” various components are also contemplated as “comprising” or “consisting essentially of” the recited components; and embodiments in the specification that recite “consisting essentially of” various components are also contemplated as “consisting of” or “comprising” the recited components (this interchangeability does not apply to the use of these terms in the claims). “Consisting essentially of” means that additional component(s), composition(s) or method step(s) that do not materially change the basic and novel characteristics of the compositions and methods described herein may be included in those compositions or methods. Such characteristics include the ability to detect an HPIV nucleic acid sequence present in a sample with specificity that distinguishes the HPIV nucleic acid from other known respiratory pathogens, optionally at a sensitivity that can detect about 1-100 TCID₅₀/ml copies of the virus, and, optionally within about 60 minutes and/or within about 40 cycles from the beginning of an amplification reaction when a cycled amplification reaction is used.

(4) A “sample” or “specimen,” including “biological” or “clinical” samples, refers to a tissue or material derived from a living or dead human or animal which may contain a parainfluenza virus target nucleic acid, including, for example, nasopharyngeal or throat swabs, nasal or bronchial washes, nasal aspirates, sputum, other respiratory tissue or exudates, biopsy tissue including lymph nodes, or body fluids such as blood or urine. A sample may be treated to physically or mechanically disrupt tissue or cell structure to release intracellular nucleic acids into a solution which may contain enzymes, buffers, salts, detergents and the like, to prepare the sample for analysis.

(5) “Nucleic acid” and “polynucleotide” refer to a multimeric compound comprising nucleosides or nucleoside analogs which have nitrogenous heterocyclic bases or base analogs linked together to form a polynucleotide, including conventional RNA, DNA, mixed RNA-DNA, and polymers that are analogs thereof. A nucleic acid “backbone” may be made up of a variety of linkages, including one or more of sugar-phosphodiester linkages, peptide-nucleic acid bonds (“peptide nucleic acids” or PNA; PCT No. WO 95/32305), phosphorothioate linkages, methylphosphonate linkages, or combinations thereof. Sugar moieties of a nucleic acid may be ribose, deoxyribose, or similar compounds with substitutions, e.g., 2' methoxy or 2' halide substitutions. Nitrogenous bases may be conventional bases (A, G, C, T, U), analogs thereof (e.g., inosine or others; see *The Biochemistry of the Nucleic Acids* 5-36, Adams et al., ed., 11th ed., 1992), derivatives of purines or pyrimidines (e.g., N⁴-methyl deoxyguanosine, deaza- or aza-purines, deaza- or aza-pyrimidines, pyrimidine bases with substituent groups at the 5 or 6 position, purine bases with a substituent at the 2, 6, or 8 positions, 2-amino-6-methylaminopurine, O⁶-methylguanine, 4-thio-pyrimidines, 4-amino-pyrimidines, 4-dimethylhydrazine-pyrimidines, and O⁴-alkyl-pyrimidines; U.S. Pat. No. 5,378,825 and PCT No. WO 93/13121). Nucleic acids may include one or more “abasic” residues where the backbone includes no nitrogenous base for position(s) of the polymer (U.S. Pat. No. 5,585,481). A nucleic acid may comprise only conventional RNA or DNA sugars, bases and linkages, or may include both conventional components and substitutions (e.g., conventional bases with 2' methoxy linkages, or polymers containing both conventional bases and one or more base analogs). Nucleic acid includes “locked nucleic acid” (LNA), an analogue containing one or more LNA nucleotide monomers with a bicyclic furanose unit locked in an RNA mimicking sugar conformation, which enhance hybridization affinity toward complementary RNA and DNA sequences (Vester and Wengel, 2004, *Biochemistry* 43(42):13233-41). Embodiments of oligomers that may affect stability of a hybridization complex include PNA oligomers, oligomers that include 2'-methoxy or 2'-fluoro substituted RNA, or oligomers that affect the overall charge, charge density, or steric associations of a hybridization complex, including oligomers that contain charged linkages (e.g., phosphorothioates) or neutral groups (e.g., methylphosphonates). 5-methylcytosines may be used in conjunction with any of the foregoing backbones/sugars/linkages including RNA or DNA backbones (or mixtures thereof) unless otherwise indicated. It is understood that when referring to ranges for the length of an oligonucleotide, amplicon, or other nucleic acid, that the range is inclusive of all whole numbers (e.g., 19-25 contiguous nucleotides in length includes 19, 20, 21, 22, 23, 24, and 25).

(6) “Antisense,” “negative-sense,” “negative-strand” HPIV nucleic acid refers to the sequence of HPIV present in infectious viruses (or a fragment thereof). “Sense,” “positive-sense,” or “positive-strand” HPIV nucleic acid is the complement of the sequence of HPIV present in infectious viruses (or a fragment thereof). Exemplary portions of HPIV-1, 2, 3, and 4 are provided in the Sequence Table below (for brevity, complete HPIV genomes, which are known in the art, are not included). Positions or nucleotides of HPIV sequences are defined with reference to the following positions of SEQ ID NO: 1-4, respectively. The first nucleotide of SEQ ID NO: 1 is HPIV-1 position 201. The first nucleotide of SEQ ID NO: 2 is HPIV-2 position 1401. The first nucleotide of SEQ ID NO: 3 is HPIV-3 position 1101. The first nucleotide of SEQ ID NO: 4 is HPIV-4 position 501. Unless otherwise indicated, “hybridizing to an HPIV nucleic acid” includes hybridizing to either a sense or antisense strand of HPIV nucleic acid, e.g., the genomic (antisense) strand, the sense transcript, or either strand of a dsDNA HPIV sequence.

(7) “C residues” include methylated and unmethylated cytosines unless the context indicates otherwise.

(8) By “RNA and DNA equivalents” is meant RNA and DNA molecules having essentially the same complementary base pair hybridization properties. RNA and DNA equivalents have different sugar moieties (i.e., ribose versus deoxyribose) and may differ by the presence of uracil in RNA and thymine in DNA. The differences between RNA and DNA

equivalents do not contribute to differences in homology because the equivalents have the same degree of complementarity to a particular sequence. Unless otherwise indicated, reference to an HPIV nucleic acid includes HPIV RNA and DNA equivalents thereof.

(9) An “oligomer” or “oligonucleotide” refers to a nucleic acid of generally less than 1,000 nucleotides (nt), including those in a size range having a lower limit of about 2 to 5 nt and an upper limit of about 500 to 900 nt. Some particular embodiments are oligomers in a size range with a lower limit of about 5 to 15, 16, 17, 18, 19, or 20 nt and an upper limit of about 50 to 600 nt, and other particular embodiments are in a size range with a lower limit of about 10 to 20 nt and an upper limit of about 22 to 100 nt. Oligomers may be purified from naturally occurring sources, but may be synthesized by using any well known enzymatic or chemical method. Oligomers may be referred to by a functional name (e.g., capture probe, primer or promoter primer) but those skilled in the art will understand that such terms refer to oligomers.

(10) By “amplicon” or “amplification product” is meant a nucleic acid molecule generated in a nucleic acid amplification reaction and which is derived from a target nucleic acid. An amplicon or amplification product contains a target nucleic acid sequence that may be of the same or opposite sense as the target nucleic acid.

(11) An “amplification oligonucleotide” or “amplification oligomer” refers to an oligonucleotide that hybridizes to a target nucleic acid, or its complement, and participates in a nucleic acid amplification reaction, e.g., serving as a primer or and promoter-primer. Particular amplification oligomers contain at least about 10 contiguous bases, and optionally at least 11, 12, 13, 14, 15, 16, 17, 18, 19, or 20 contiguous bases, that are complementary to a region of the target nucleic acid sequence or its complementary strand. The contiguous bases may be at least about 80%, at least about 90%, or completely complementary to the target sequence to which the amplification oligomer binds. One skilled in the art will understand that the recited ranges include all whole and rational numbers within the range (e.g., 92% or 98.377%). Particular amplification oligomers are about 10 to about 60 bases long and optionally may include modified nucleotides.

(12) A “primer” refers to an oligomer that hybridizes to a template nucleic acid and has a 3' end that is extended by polymerization. A primer may be optionally modified, e.g., by including a 5' region that is non-complementary to the target sequence. Such modification can include functional additions, such as tags, promoters, or other sequences used or useful for manipulating or amplifying the primer or target oligonucleotide.

(13) Within the context of transcription mediated amplification, a primer modified with a 5' promoter sequence may be referred to as a “promoter-primer.” A person of ordinary skill in the art of molecular biology or biochemistry will understand that an oligomer that can function as a primer can be modified to include a 5' promoter sequence and then function as a promoter-primer, and, similarly, any promoter-primer can serve as a primer with or without its 5' promoter sequence.

(14) “Nucleic acid amplification” refers to any in vitro procedure that produces multiple copies of a target nucleic acid sequence, or its complementary sequence, or fragments thereof (i.e., an amplified sequence containing less than the complete target nucleic acid). Examples of nucleic acid amplification procedures include transcription associated methods, such as transcription-mediated amplification (TMA), nucleic acid sequence-based amplification (NASBA) and others (e.g., U.S. Pat. Nos. 5,399,491, 5,554,516, 5,437,990, 5,130,238, 4,868,105, and 5,124,246), replicase-mediated amplification (e.g., U.S. Pat. No. 4,786,600), the polymerase chain reaction (PCR) (e.g., U.S. Pat. Nos. 4,683,195, 4,683,202, and 4,800,159), ligase chain reaction (LCR) (e.g., EP Pat. App. 0320308), helicase-dependent amplification (e.g., U.S. Pat. No. 7,282,328), and strand-displacement amplification (SDA) (e.g., U.S. Pat. No. 5,422,252).

Amplification may be linear or exponential. Replicase-mediated amplification uses self-replicating RNA molecules, and a replicase such as QB-replicase. PCR amplification uses DNA polymerase, primers, and thermal cycling steps to synthesize multiple copies of the two complementary strands of DNA or cDNA. LCR amplification uses at least four separate oligonucleotides to amplify a target and its complementary strand by using multiple cycles of hybridization, ligation, and denaturation. Helicase-dependent amplification uses a helicase to separate the two strands of a DNA duplex generating single-stranded templates, followed by hybridization of sequence-specific primers hybridize to the templates and extension by DNA polymerase to amplify the target sequence. SDA uses a primer that contains a recognition site for a restriction endonuclease that will nick one strand of a hemimodified DNA duplex that includes the target sequence, followed by amplification in a series of primer extension and strand displacement steps. Particular embodiments use PCR or TMA, but it will be apparent to persons of ordinary skill in the art that oligomers disclosed herein may be readily used as primers in other amplification methods.

(15) Transcription associated amplification uses a DNA polymerase, an RNA polymerase, deoxyribonucleoside triphosphates, ribonucleoside triphosphates, a promoter-containing oligonucleotide, and optionally may include other oligonucleotides, to ultimately produce multiple RNA transcripts from a nucleic acid template (described in detail in U.S. Pat. Nos. 5,399,491 and 5,554,516, Kacian et al., U.S. Pat. No. 5,437,990, Burg et al., PCT Nos. WO 88/01302 and WO 88/10315, Gingeras et al., U.S. Pat. No. 5,130,238, Malek et al., U.S. Pat. Nos. 4,868,105 and 5,124,246, Urdea et al., PCT No. WO 94/03472, McDonough et al., PCT No. WO 95/03430, and Ryder et al.). Methods that use TMA are described in detail previously (U.S. Pat. Nos. 5,399,491 and 5,554,516).

(16) In cyclic amplification methods that detect amplicons in real-time, the term “Threshold cycle” (Ct) is a measure of the emergence time of a signal associated with amplification of target, and is generally 10× standard deviation of the normalized reporter signal. Once an amplification reaches the “threshold cycle,” generally there is considered to be a positive amplification product of a sequence to which the probe binds. The identity of the amplification product can then be determined through methods known to one of skill in the art, such as gel electrophoresis, nucleic acid sequencing, and

other such analytical procedures.

(17) As used herein, the term “relative fluorescence unit” (“RFU”) is a unit of measurement of fluorescence intensity. RFU varies with the characteristics of the detection means used for the measurement, and can be used as a measurement to compare relative intensities between samples and controls. The analytical sensitivity (limit of detection or LoD) is determined from the median tissue culture infective dose (TCID₅₀/ml). The TCID₅₀/ml is that amount of a pathogenic agent that will produce pathological change in 50% of cell cultures inoculated.

(18) “Detection probe” or “probe” refers to an oligomer that hybridizes specifically to a target sequence, including an amplified sequence, under conditions that promote nucleic acid hybridization, for detection of the target nucleic acid. Detection may either be direct (i.e., probe hybridized directly to the target) or indirect (i.e., a probe hybridized to an intermediate structure that links the probe to the target). A probe's target sequence generally refers to the specific sequence within a larger sequence which the probe hybridizes specifically. A detection probe may include target-specific sequences and a non-target-complementary sequence. Such non-target-complementary sequences can include sequences which will confer a desired secondary or tertiary structure, such as a hairpin structure, which can be used to facilitate detection and/or amplification (e.g., U.S. Pat. Nos. 5,118,801, 5,312,728, 6,835,542, and 6,849,412). Probes of a defined sequence may be produced by techniques known to those of ordinary skill in the art, such as by chemical synthesis, and by in vitro or in vivo expression from recombinant nucleic acid molecules.

(19) By “hybridization” or “hybridize” is meant the ability of two completely or partially complementary nucleic acid strands to come together under specified hybridization assay conditions in a parallel or antiparallel orientation to form a stable structure having a double-stranded region. The two constituent strands of this double-stranded structure, sometimes called a hybrid, are held together by hydrogen bonds. Although these hydrogen bonds most commonly form between nucleotides containing the bases adenine and thymine or uracil (A and T or U) or cytosine and guanine (C and G) on single nucleic acid strands, base pairing can also form between bases which are not members of these “canonical” pairs. Non-canonical base pairing is well-known in the art. (See, e.g., R. L. P. Adams et al., *The Biochemistry of the Nucleic Acids* (11th ed. 1992).)

(20) By “preferentially hybridize” is meant that under stringent hybridization conditions, an amplification or detection probe oligomer can hybridize to its target nucleic acid to form stable oligomer:target hybrid, but not form a sufficient number of stable oligomer:non-target hybrids. Amplification and detection oligomers that preferentially hybridize to a target nucleic acid are useful to amplify and detect target nucleic acids, but not non-targeted organisms, especially phylogenetically closely related organisms. Thus, the oligomer hybridizes to target nucleic acid to a sufficiently greater extent than to non-target nucleic acid to enable one having ordinary skill in the art to accurately amplify and/or detect the presence (or absence) of nucleic acid derived from the specified influenza viruses as appropriate. In general, reducing the degree of complementarity between an oligonucleotide sequence and its target sequence will decrease the degree or rate of hybridization of the oligonucleotide to its target region. However, the inclusion of one or more non-complementary nucleosides or nucleobases may facilitate the ability of an oligonucleotide to discriminate against non-target organisms.

(21) Preferential hybridization can be measured using techniques known in the art and described herein, such as in the examples provided below. In some embodiments, there is at least a 10-fold difference between target and non-target hybridization signals in a test sample, at least a 100-fold difference, or at least a 1,000-fold difference. In some embodiments, non-target hybridization signals in a test sample are no more than the background signal level.

(22) By “stringent hybridization conditions,” or “stringent conditions” is meant conditions permitting an oligomer to preferentially hybridize to a target nucleic acid (such as an HPIV nucleic acid) and not to nucleic acid derived from a closely related non-target nucleic acid. While the definition of stringent hybridization conditions does not vary, the actual reaction environment that can be used for stringent hybridization may vary depending upon factors including the GC content and length of the oligomer, the degree of similarity between the oligomer sequence and sequences of non-target nucleic acids that may be present in the test sample, and the target sequence. Hybridization conditions include the temperature and the composition of the hybridization reagents or solutions. Exemplary hybridization assay conditions for amplifying and/or detecting target nucleic acids derived from one or more strains of HPIV with the oligomers of the present disclosure correspond to a temperature of about 60° C. when the salt concentration, such as a monovalent salt, e.g., KCl, is in the range of about 0.6-0.9 M. Specific hybridization assay conditions are set forth infra in the Examples section. Other acceptable stringent hybridization conditions could be easily ascertained by those having ordinary skill in the art.

(23) By “competes for hybridization to an HPIV nucleic acid under stringent conditions” with a referenced oligomer is meant that an oligomer substantially reduces the binding of the referenced oligomer to its target HPIV sequence under stringent conditions, the competing oligomer when supplied in excess can reduce binding of the referenced oligomer at a sub-saturating concentration by about 20%, 30%, 40%, 50%, or more, or the T_m of the competing oligomer is higher than or within about 5, 4, 3, 2, or 1° C. of the T_m of the referenced oligomer to the target. Suitable oligonucleotide competition assay conditions and procedures are known in the art.

(24) By “assay conditions” is meant conditions permitting stable hybridization of an oligonucleotide to a target nucleic acid. Assay conditions do not require preferential hybridization of the oligonucleotide to the target nucleic acid.

(25) “Label” or “detectable label” refers to a moiety or compound joined directly or indirectly to a probe that is detected or leads to a detectable signal. Direct joining may use covalent bonds or non-covalent interactions (e.g., hydrogen bonding, hydrophobic or ionic interactions, and chelate or coordination complex formation) whereas indirect joining may

use a bridging moiety or linker (e.g., via an antibody or additional oligonucleotide(s), which amplify a detectable signal. Any detectable moiety may be used, e.g., radionuclide, ligand such as biotin or avidin, enzyme, enzyme substrate, reactive group, chromophore such as a dye or particle (e.g., latex or metal bead) that imparts a detectable color, luminescent compound (e.g. bioluminescent, phosphorescent, or chemiluminescent compound), and fluorescent compound (i.e., fluorophore). Embodiments of fluorophores include those that absorb light in the range of about 495 to 650 nm and emit light in the range of about 520 to 670 nm, which include those known as FAM™, TET™, CAL FLUOR™ (Orange or Red), and QUASAR™ compounds. Fluorophores may be used in combination with a quencher molecule that absorbs light when in close proximity to the fluorophore to diminish background fluorescence. Such quenchers are well known in the art and include, e.g., BLACK HOLE QUENCHER™ (or BHQ™) or TAMRA™ compounds. Particular embodiments include a “homogeneous detectable label” that is detectable in a homogeneous system in which bound labeled probe in a mixture exhibits a detectable change compared to unbound labeled probe, which allows the label to be detected without physically removing hybridized from unhybridized labeled probe (e.g., U.S. Pat. Nos. 5,283,174, 5,656,207, and 5,658,737). Particular homogeneous detectable labels include chemiluminescent compounds, including acridinium ester (“AE”) compounds, such as standard AE or AE derivatives which are well known (U.S. Pat. Nos. 5,656,207, 5,658,737, and 5,639,604). Methods of synthesizing labels, attaching labels to nucleic acid, and detecting signals from labels are well known (e.g., Sambrook et al., *Molecular Cloning, A Laboratory Manual*, 2nd ed. (Cold Spring Harbor Laboratory Press, Cold Spring Harbor, NY, 1989) at Chapt. 10, and U.S. Pat. Nos. 5,658,737, 5,656,207, 5,547,842, 5,283,174, and 4,581,333, and EP Pat. App. 0 747 706). Particular methods of linking an AE compound to a nucleic acid are known (e.g., U.S. Pat. Nos. 5,585,481 and 5,639,604, see column 10, line 6 to column 11, line 3, and Example 8). Particular AE labeling positions are a probe's central region and near a region of A/T base pairs, at a probe's 3' or 5' terminus, or at or near a mismatch site with a known sequence that is the probe should not detect compared to the desired target sequence. Other detectably labeled probes include TaqMan™ probes, molecular torches, and molecular beacons. TaqMan™ probes include a donor and acceptor label wherein fluorescence is detected upon enzymatically degrading the probe during amplification in order to release the fluorophore from the presence of the quencher. Molecular torches and beacons exist in open and closed configurations wherein the closed configuration quenches the fluorophore and the open position separates the fluorophore from the quencher to allow fluorescence. Hybridization to target opens the otherwise closed probes.

(26) Sequences are “sufficiently complementary” if they allow stable hybridization of two nucleic acid sequences, e.g., stable hybrids of probe and target sequences, although the sequences need not be completely complementary. That is, a “sufficiently complementary” sequence that hybridizes to another sequence by hydrogen bonding between a subset series of complementary nucleotides by using standard base pairing (e.g., G:C, A:T, or A:U), although the two sequences may contain one or more residues (including abasic positions) that are not complementary so long as the entire sequences in appropriate hybridization conditions to form a stable hybridization complex. Sufficiently complementary sequences may be at least about 80%, at least about 90%, or completely complementary in the sequences that hybridize together. Appropriate hybridization conditions are well known to those skilled in the art, can be predicted based on sequence composition, or can be determined empirically by using routine testing (e.g., Sambrook et al., *Molecular Cloning, A Laboratory Manual*, 2^{sup}.nd ed. at  custom character 1.90-1.91, 7.37-7.57, 9.47-9.51 and 11.47-11.57, particularly  custom character 9.50-9.51, 11.12-11.13, 11.45-11.47 and 11.55-11.57).

(27) A “non-extendable” oligomer includes a blocking moiety at or near its 3'-terminus to prevent extension. A blocking group near the 3' end is in some embodiments within five residues of the 3' end and is sufficiently large to limit binding of a polymerase to the oligomer, and other embodiments contain a blocking group covalently attached to the 3' terminus. Many different chemical groups may be used to block the 3' end, e.g., alkyl groups, non-nucleotide linkers, alkane-diol dideoxynucleotide residues, and cordycepin. Further examples of blocking moieties include a 3'-deoxy nucleotide (e.g., a 2',3'-dideoxy nucleotide); a 3'-phosphorylated nucleotide; a fluorophore, quencher, or other label that interferes with extension; an inverted nucleotide (e.g., linked to the preceding nucleotide through a 3'-to-3' phosphodiester, optionally with an exposed 5'-OH or phosphate); or a protein or peptide joined to the oligonucleotide so as to prevent further extension of a nascent nucleic acid chain by a polymerase. A non-extendable oligonucleotide of the present disclosure may be at least 10 bases in length, and may be up to 15, 20, 25, 30, 35, 40, 50 or more nucleotides in length. Non-extendable oligonucleotides that comprise a detectable label can be used as probes.

(28) References, particularly in the claims, to “the sequence of SEQ ID NO: X” refer to the base sequence of the corresponding sequence listing entry and do not require identity of the backbone (e.g., RNA, 2'-O-Me RNA, or DNA) or base modifications (e.g., methylation of cytosine residues) unless otherwise indicated.

(29) A “degenerate” position in an oligomer refers to a position where more than one base pair is present in a population of the oligomer. For example, in SEQ ID NO: 5, the third nucleotide is Y, which represents C or T/U. Oligomers with degenerate positions can be synthesized by providing a mixture of nucleotide precursors corresponding to the desired degenerate combination at the step of the synthesis where incorporation of a degenerate position is desired.

(30) A “non-Watson Crick” (NWC) position in an oligomer refers to a position where the oligomer is configured to hybridize to at least one HPIV target sequence with a non-Watson Crick pairing, such as G-U, G-T, or G-A (either the G or the U/T/A can be the base in the oligomer). In some embodiments, the NWC position is configured to hybridize via a wobble (G-U) or purine-purine (G-A) pair. In some embodiments, the NWC position is configured to hybridize via a G-T pair. In some embodiments, the NWC position is configured to hybridize via a G-U pair. In some embodiments, the NWC

position is configured to hybridize via a G-A pair.

(31) Unless defined otherwise, all scientific and technical terms used herein have the same meaning as commonly understood by those skilled in the relevant art. General definitions may be found in technical books relevant to the art of molecular biology, e.g., *DICTIONARY OF MICROBIOLOGY AND MOLECULAR BIOLOGY*, 2nd ed. (Singleton et al., 1994, John Wiley & Sons, New York, NY) or *THE HARPER COLLINS DICTIONARY OF BIOLOGY* (Hale & Marham, 1991, Harper Perennial, New York, NY).

Exemplary Compositions, Kits, Methods, and Uses

(32) The present disclosure provides oligomers, compositions, and kits, useful for amplifying, detecting, or quantifying HPIV-1, 2, 3, and/or 4 from a sample.

(33) In some embodiments, oligomers are provided, e.g., in a kit or composition. Oligomers generally comprise a target-hybridizing region, e.g., configured to hybridize specifically to an HPIV nucleic acid. While oligomers of different lengths and base composition may be used for amplifying HPIV nucleic acids, in some embodiments oligomers in this disclosure have target-hybridizing regions from about 10-60 bases in length, about 14-50 bases in length, about 14-40 bases in length, about 14-35 bases in length, or about 15-30 bases in length. In some embodiments, an oligomer comprises a second region of sequence in addition to the target-hybridizing region, such as a promoter, which can be located 5' of the target-hybridizing region. In some embodiments, an oligomer does not comprise a second region of sequence.

(34) In some embodiments, a pair of oligomers is provided wherein one oligomer is configured to hybridize to a sense strand of an HPIV nucleic acid and the other is configured to hybridize to an anti-sense strand of an HPIV nucleic acid. Such oligomers include primer pairs for PCR or other forms of amplification.

(35) In some embodiments, one or more oligomers, such as a primer pair or a primer pair and a third oligomer which is optionally labeled (e.g., for use as a probe), are configured to hybridize to an HPIV hemagglutinin-neuraminidase (HN) gene. In some embodiments, the HN gene is an HPIV-1, 2, or 3 HN gene. In some embodiments, a plurality of oligomers, such as a plurality of primer pairs or a plurality of primer pairs and third oligomers which are optionally labeled (e.g., for use as a probe), are provided which collectively hybridize to the HN gene of HPIV-1 and 2; HPIV-1 and 3; HPIV-2 and 3; or HPIV-1, 2, and 3.

(36) In some embodiments, one or more oligomers, such as a primer pair or a primer pair and a third oligomer which is optionally labeled (e.g., for use as a probe), are configured to hybridize to an HPIV nucleocapsid (NP) gene. In some embodiments, the NP gene is an HPIV-4 NP gene, such as an HPIV-4a or 4b NP gene. In some embodiments, the one or more oligomers are configured to hybridize to either or both of an HPIV-4a or 4b NP gene.

(37) In some embodiments, one or more oligomers comprise a degenerate position. In some embodiments, an HPIV-1 primer, an HPIV-1 primer pair, and/or an HPIV-1 probe comprises a degenerate position. In some embodiments, an HPIV-3 primer, an HPIV-3 primer pair, and/or an HPIV-3 probe comprises a degenerate position. In some embodiments, an HPIV-4 primer, an HPIV-4 primer pair, and/or an HPIV-4 probe comprises a degenerate position. In some embodiments, the degenerate position is Y, R, S, W, K, or M. Exemplary oligomers with degenerate positions (e.g., Y, R, S, W, K, or M) are listed in the Sequence Table below.

(38) In some embodiments, one or more oligomers comprise a non-Watson Crick (NWC) position. In some embodiments, an HPIV-1 primer, an HPIV-1 primer pair, and/or an HPIV-1 probe comprises a NWC position. In some embodiments, an HPIV-2 primer, an HPIV-2 primer pair, and/or an HPIV-2 probe comprises a degenerate position. In some embodiments, an HPIV-2 primer, an HPIV-2 primer pair, and/or an HPIV-2 probe comprises a NWC position. In some embodiments, an HPIV-3 primer, an HPIV-3 primer pair, and/or an HPIV-3 probe comprises a NWC position. In some embodiments, an HPIV-4 primer, an HPIV-4 primer pair, and/or an HPIV-4 probe comprises a NWC position.

(39) Exemplary NWC positions include U residues in various exemplary oligomers in the Sequence Table. Additional exemplary NWC positions include the G at position 11 of SEQ ID NO: 38; and the G at position 8 of SEQ ID NO: 78.

(40) In some embodiments, one or more oligomers in a set, kit, composition, or reaction mixture comprise a methylated cytosine (e.g., 5-methylcytosine). In some embodiments, at least about half of the cytosines in an oligomer are methylated. In some embodiments, all or substantially all (e.g., all but one or two) of the cytosines in an oligomer are methylated. In some embodiments, a cytosine at the 3' end or within 2, 3, 4, or 5 bases of the 3' end is unmethylated.

(41) Exemplary primer pairs and optional third oligomers are set forth in the following table.

(42) TABLE-US-00001 TABLE A Exemplary oligomer sets. Oligomers are referred to by their SEQ ID NO (see the Sequence Table below). Oligomer 1 Oligomer 2 (e.g., forward (e.g., reverse Oligomer 3 (optionally labeled, e.g., primer) primer) probe) HPIV-1 6 24 10 or 11 17 22 5, 7, 8, 9, 12, 13, 14, 15, 16, 18, 19, 20, 21, 23, or 125 HPIV-2 44 40 38, 39, 41, 42, 43, or 45 HPIV-3 25 36 26, 27, 28, 29, or 30 28 37 31, 32, 33, 34, or 35 HPIV-4 72 50, 51, 52, 46, 47, 48, 49, 59, 61, 69, 70, 72, 73, 74, 93, 95, or 96 75, 77, 88, 90, 91, 116, 117, 118, 119, 121, 122, or 123 78 56 71 79, 80, 81, 54 53, 55, 58, 60, 62, 65, 67, 68, 76, 83, 84, 85, 82, 97, 98, 86, 92, 94, 101, 102, 103, 104, 105, 106, 99, or 100 107, 108, 109, 110, 111, or 112 79, 80, 81, 63 53, 55, 58, 60, 62, 65, 67, 68, 76, 83, 84, 85, 82, 97, 98, 86, 92, 94, 101, 102, 103, 104, 105, 106, 99, or 100 107, 108, 109, 110, 111, or 112 79, 80, 81, 64 53, 55, 58, 60, 62, 65, 67, 68, 76, 83, 84, 85, 82, 97, 98, 86, 92, 94, 101, 102, 103, 104, 105, 106, 99, or 100 107, 108, 109, 110, 111, or 112 79, 80, 81, 113 53, 55, 58, 60, 62, 65, 67, 68, 76, 83, 84, 85, 82, 97, 98, 86, 92, 94, 101, 102, 103, 104, 105, 106, 99, or 100 107, 108, 109, 110, 111, or 112 79, 80, 81, 114 53, 55, 58, 60, 62, 65, 67, 68, 76, 83, 84, 85, 82, 97, 98, 86, 92, 94, 101, 102, 103, 104, 105, 106, 99, or 100 107, 108, 109, 110, 111, or 112 79, 80, 81, 115 53, 55, 58, 60, 62, 65, 67, 68, 76, 83, 84, 85, 82, 97, 98, 86, 92, 94, 101,

102, 103, 104, 105, 106, 99, or 100 107, 108, 109, 110, 111, or 112 124 50, 51, 52, 46, 47, 48, 49, 59, 61, 69, 70, 72, 73, 74, 75, 93, 95, or 96 77, 88, 90, 91, 116, 117, 118, 119, 121, 122, or 123

(43) In some embodiments, an oligomer is provided that comprises a label. Such an oligomer can be used as a probe. In some embodiments, the labeled oligomer has a sequence corresponding to a SEQ ID NO listed in the Oligomer 3 column of Table A. In some embodiments, the label is a non-nucleotide label. Suitable labels include compounds that emit a detectable light signal, e.g., fluorophores or luminescent (e.g., chemiluminescent) compounds that can be detected in a homogeneous mixture. More than one label, and more than one type of label, may be present on a particular probe, or detection may rely on using a mixture of probes in which each probe is labeled with a compound that produces a detectable signal (see, e.g., U.S. Pat. Nos. 6,180,340 and 6,350,579, each incorporated by reference herein). Labels may be attached to a probe by various means including covalent linkages, chelation, and ionic interactions, but in some embodiments the label is covalently attached. For example, in some embodiments, a detection probe has an attached chemiluminescent label such as, e.g., an acridinium ester (AE) compound (see, e.g., U.S. Pat. Nos. 5,185,439; 5,639,604; 5,585,481; and 5,656,744). A label, such as a fluorescent or chemiluminescent label, can be attached to the probe by a non-nucleotide linker (see, e.g., U.S. Pat. Nos. 5,585,481; 5,656,744; and 5,639,604).

(44) In some embodiments, a probe (e.g., comprising a fluorescent label) further comprises a second label that interacts with the first label. For example, the second label can be a quencher. Such probes can be used, e.g., in TaqMan™ assays, where hybridization of the probe to a target or amplicon followed by nucleolysis by a polymerase comprising 5'-3' exonuclease activity results in liberation of the fluorescent label and thereby increased fluorescence, or fluorescence independent of the interaction with the second label.

(45) In some applications, one or more probes exhibiting at least some degree of self-complementarity are used to facilitate detection of probe:target duplexes in a test sample without first requiring the removal of unhybridized probe prior to detection. Specific embodiments of such detection probes include, for example, probes that form conformations held by intramolecular hybridization, such as conformations generally referred to as hairpins. Suitable hairpin probes include a "molecular torch" (see, e.g., U.S. Pat. Nos. 6,849,412; 6,835,542; 6,534,274; and 6,361,945) and a "molecular beacon" (see, e.g., U.S. Pat. Nos. 5,118,801 and 5,312,728). Molecular torches include distinct regions of self-complementarity (coined "the target binding domain" and "the target closing domain") which are connected by a joining region (e.g., a —(CH.sub.2CH.sub.2O).sub.3— linker) and which hybridize to one another under predetermined hybridization assay conditions. When exposed to an appropriate target or denaturing conditions, the two complementary regions (which may be fully or partially complementary) of the molecular torch melt, leaving the target binding domain available for hybridization to a target sequence when the predetermined hybridization assay conditions are restored. Molecular torches are designed so that the target binding domain favors hybridization to the target sequence over the target closing domain. The target binding domain and the target closing domain of a molecular torch include interacting labels (e.g., fluorescent/quencher) positioned so that a different signal is produced when the molecular torch is self-hybridized as opposed to when the molecular torch is hybridized to a target nucleic acid, thereby permitting detection of probe:target duplexes in a test sample in the presence of unhybridized probe having a viable label associated therewith.

(46) Examples of interacting donor/acceptor label pairs that may be used in connection with the disclosure, making no attempt to distinguish FRET from non-FRET pairs, include fluorescein/tetramethylrhodamine, IAEDANS/fluorescein, EDANS/DABCYL, coumarin/DABCYL, fluorescein/fluorescein, BODIPY FL/BODIPY FL, fluorescein/DABCYL, lucifer yellow/DABCYL, BODIPY/DABCYL, eosine/DABCYL, erythrosine/DABCYL, tetramethylrhodamine/DABCYL, Texas Red/DABCYL, CY5/BH1, CY5/BH2, CY3/BH1, CY3/BH2 and fluorescein/QSY7 dye. Those having an ordinary level of skill in the art will understand that when donor and acceptor dyes are different, energy transfer can be detected by the appearance of sensitized fluorescence of the acceptor or by quenching of donor fluorescence. Non-fluorescent acceptors such as DABCYL and the QSY7 dyes advantageously eliminate the potential problem of background fluorescence resulting from direct (i.e., non-sensitized) acceptor excitation. Exemplary fluorophore moieties that can be used as one member of a donor-acceptor pair include fluorescein, ROX, and the CY dyes (such as CY5). Exemplary quencher moieties that can be used as another member of a donor-acceptor pair include DABCYL and the BLACK HOLE QUENCHER moieties which are available from Biosearch Technologies, Inc., (Novato, Calif.).

(47) In some embodiments, a labeled oligomer (e.g., probe) is non-extendable. For example, the labeled oligomer can be rendered non-extendable by 3'-phosphorylation, having a 3'-terminal 3'-deoxynucleotide (e.g., a terminal 2',3'-dideoxynucleotide), having a 3'-terminal inverted nucleotide (e.g., in which the last nucleotide is inverted such that it is joined to the penultimate nucleotide by a 3' to 3' phosphodiester linkage or analog thereof, such as a phosphorothioate), or having an attached fluorophore, quencher, or other label that interferes with extension (possibly but not necessarily attached via the 3' position of the terminal nucleotide). In some embodiments, the 3'-terminal nucleotide is not methylated.

(48) Also provided by the disclosure is a reaction mixture for determining the presence or absence of an HPIV target nucleic acid or quantifying the amount thereof in a sample. A reaction mixture in accordance with the present disclosure comprises at least one or more of the following: an oligomer combination as described herein for amplification of an HPIV target nucleic acid; and a detection probe oligomer as described herein for determining the presence or absence of an HPIV amplification product. The reaction mixture may further include a number of optional components such as, for example, capture probes, e.g., poly-(k) capture probes as described in US 2013/0209992, which is incorporated herein by

reference. For an amplification reaction mixture, the reaction mixture will typically include other reagents suitable for performing in vitro amplification such as, e.g., buffers, salt solutions, appropriate nucleotide triphosphates (e.g., dATP, dCTP, dGTP, and dTTP; and/or ATP, CTP, GTP and UTP), and/or enzymes (e.g., a thermostable DNA polymerase, or reverse transcriptase and/or RNA polymerase), and will typically include test sample components, in which an HPIV target nucleic acid may or may not be present. A reaction mixture may include amplification oligomers for only one target region of an HPIV genome, or it may include amplification oligomers for multiple HPIV target regions of the same or different type or subtypes, e.g., one, two, three, or four of HPIV-1, HPIV-2, HPIV-3, and HPIV-4. In addition, for a reaction mixture that includes a detection probe together with an amplification oligomer combination, selection of amplification oligomers and detection probe oligomers for a reaction mixture are linked by a common target region (i.e., the reaction mixture will include a probe that binds to a sequence amplifiable by an amplification oligomer combination of the reaction mixture).

(49) In some embodiments, the reaction mixture comprises KCl. In some embodiments, the KCl concentration is about 50 mM. In some embodiments, the KCl concentration is greater than about 50 mM, e.g., about 60-150 mM, about 75-125 mM, about 80-120 mM, about 85-115 mM, or about 90-110 mM. In some embodiments, the KCl concentration is 55-65, 65-75, 75-85, 85-95, 95-105, 105-115, 115-125, 125-135, or 135-145, wherein each of the foregoing is in mM and is optionally modified by “about”. In some embodiments, a composition according to the disclosure comprises KCl, e.g., at any of the foregoing concentrations. In some embodiments, a method according to the disclosure comprises performing an amplification reaction in the presence of KCl, e.g., at any of the foregoing concentrations.

(50) Also provided by the subject disclosure are kits for practicing the methods as described herein. A kit in accordance with the present disclosure comprises at least one or more of the following: an amplification oligomer combination as described herein for amplification of an HPIV target nucleic acid; and at least one detection probe oligomer as described herein for determining the presence or absence of an HPIV amplification product. In some embodiments, any oligomer combination described herein is present in the kit. The kits may further include a number of optional components such as, for example, capture probes, e.g., poly-(k) capture probes as described in US 2013/0209992. Other reagents that may be present in the kits include reagents suitable for performing in vitro amplification such as, e.g., buffers, salt solutions, appropriate nucleotide triphosphates (e.g., dATP, dCTP, dGTP, dTTP; and/or ATP, CTP, GTP and UTP), and/or enzymes (e.g., a thermostable DNA polymerase, or a reverse transcriptase and/or RNA polymerase). Oligomers as described herein may be packaged in a variety of different embodiments, and those skilled in the art will appreciate that the disclosure embraces many different kit configurations. For example, a kit may include amplification oligomers for only one target region of an HPIV genome, or it may include amplification oligomers for multiple HPIV target regions of the same or different type or subtypes, e.g., one, two, three, or four of HPIV-1, HPIV-2, HPIV-3, and HPIV-4. In addition, for a kit that includes a detection probe together with an amplification oligomer combination, selection of amplification oligomers and detection probe oligomers for a kit are linked by a common target region (i.e., the kit will include a probe that binds to a sequence amplifiable by an amplification oligomer combination of the kit). In certain embodiments, the kit further includes a set of instructions for practicing methods in accordance with the present disclosure, where the instructions may be associated with a package insert and/or the packaging of the kit or the components thereof.

(51) Any method disclosed herein is also to be understood as a disclosure of corresponding uses of materials involved in the method directed to the purpose of the method. Any of the oligomers comprising HPIV sequence and any combinations (e.g., kits and compositions) comprising such an oligomer are to be understood as also disclosed for use in detecting or quantifying HPIV, and for use in the preparation of a composition for detecting or quantifying HPIV.

(52) Broadly speaking, methods can comprise one or more of the following components: target capture, in which HPIV nucleic acid (e.g., from a sample, such as a clinical sample) is annealed to a capture oligomer; isolation, e.g., washing, to remove material not associated with a capture oligomer; amplification; and amplicon detection, e.g., amplicon quantification, which may be performed in real time with amplification. Certain embodiments involve each of the foregoing steps. Certain embodiments involve exponential amplification, optionally with a preceding linear amplification step. Certain embodiments involve exponential amplification and amplicon detection. Certain embodiments involve any two of the components listed above. Certain embodiments involve any two components listed adjacently above, e.g., washing and amplification, or amplification and detection.

(53) In some embodiments, amplification comprises contacting the sample with at least two oligomers for amplifying an HPIV nucleic acid target region corresponding to an HPIV target nucleic acid, where the oligomers include at least two amplification oligomers as described above (e.g., one or more oriented in the sense direction and one or more oriented in the antisense direction for exponential amplification); (2) performing an in vitro nucleic acid amplification reaction, where any HPIV target nucleic acid present in the sample is used as a template for generating an amplification product; and (3) detecting the presence or absence of the amplification product, thereby determining the presence or absence of HPIV in the sample, or quantifying the amount of HPIV nucleic acid in the sample.

(54) A detection method in accordance with the present disclosure can further include the step of obtaining the sample to be subjected to subsequent steps of the method. In certain embodiments, “obtaining” a sample to be used includes, for example, receiving the sample at a testing facility or other location where one or more steps of the method are performed, and/or retrieving the sample from a location (e.g., from storage or other depository) within a facility where one or more steps of the method are performed.

(55) In certain embodiments, the method further includes purifying the HPIV target nucleic acid from other components

in the sample, e.g., before an amplification, such as before a capture step. Such purification may include methods of separating and/or concentrating organisms contained in a sample from other sample components, or removing or degrading non-nucleic acid sample components, e.g., protein, carbohydrate, salt, lipid, etc. In some embodiments, DNA in the sample is degraded, e.g., with DNase, and optionally removing or inactivating the DNase or removing degraded DNA.

(56) In particular embodiments, purifying the target nucleic acid includes capturing the target nucleic acid to specifically or non-specifically separate the target nucleic acid from other sample components. Non-specific target capture methods may involve selective precipitation of nucleic acids from a substantially aqueous mixture, adherence of nucleic acids to a support that is washed to remove other sample components, or other means of physically separating nucleic acids from a mixture that contains HPIV nucleic acid and other sample components.

(57) Target capture typically occurs in a solution phase mixture that contains one or more capture probe oligomers that hybridize to the HPIV target sequence under hybridizing conditions. For embodiments comprising a capture probe tail, the HPIV-target:capture-probe complex is captured by adjusting the hybridization conditions so that the capture probe tail hybridizes to the immobilized probe. Certain embodiments use a particulate solid support, such as paramagnetic beads.

(58) Isolation can follow capture, wherein the complex on the solid support is separated from other sample components. Isolation can be accomplished by any appropriate technique, e.g., washing a support associated with the HPIV-target-sequence one or more times (e.g., 2 or 3 times) to remove other sample components and/or unbound oligomer. In embodiments using a particulate solid support, such as paramagnetic beads, particles associated with the HPIV-target may be suspended in a washing solution and retrieved from the washing solution. In some embodiments by using magnetic attraction. To limit the number of handling steps, the HPIV target nucleic acid may be amplified by simply mixing the HPIV target sequence in the complex on the support with amplification oligomers and proceeding with amplification steps.

(59) Exponentially amplifying an HPIV target sequence utilizes an in vitro amplification reaction using at least two amplification oligomers that flank a target region to be amplified. In some embodiments, at least first and second oligomers as described above are provided. In some embodiments, a plurality of pairs of oligomers is provided, wherein the plurality comprises oligomer pairs configured to hybridize to at least two, three, or four of an HPIV-1, HPIV-2, HPIV-3, or HPIV-4 nucleic acid. The amplification reaction can be cycled or isothermal. Suitable amplification methods include, for example, replicase-mediated amplification, polymerase chain reaction (PCR), ligase chain reaction (LCR), strand-displacement amplification (SDA), and transcription-mediated or transcription-associated amplification (TMA).

(60) A detection step may be performed using any of a variety of known techniques to detect a signal specifically associated with the amplified target sequence, such as, e.g., by hybridizing the amplification product with a labeled detection probe and detecting a signal resulting from the labeled probe (including from label released from the probe following hybridization in some embodiments). In some embodiments, the labeled probe comprises a second moiety, such as a quencher or other moiety that interacts with the first label, as discussed above. The detection step may also provide additional information on the amplified sequence, such as, e.g., all or a portion of its nucleic acid base sequence.

Detection may be performed after the amplification reaction is completed, or may be performed simultaneously with amplifying the target region, e.g., in real time. In one embodiment, the detection step allows homogeneous detection, e.g., detection of the hybridized probe without removal of unhybridized probe from the mixture (see, e.g., U.S. Pat. Nos. 5,639,604 and 5,283,174). In some embodiments, the nucleic acids are associated with a surface that results in a physical change, such as a detectable electrical change. Amplified nucleic acids may be detected by concentrating them in or on a matrix and detecting the nucleic acids or dyes associated with them (e.g., an intercalating agent such as ethidium bromide or cyber green), or detecting an increase in dye associated with nucleic acid in solution phase. Other methods of detection may use nucleic acid detection probes that are configured to specifically hybridize to a sequence in the amplified product and detecting the presence of the probe:product complex, or by using a complex of probes that may amplify the detectable signal associated with the amplified products (e.g., U.S. Pat. Nos. 5,424,413; 5,451,503; and 5,849,481; each incorporated by reference herein). Directly or indirectly labeled probes that specifically associate with the amplified product provide a detectable signal that indicates the presence of the target nucleic acid in the sample. In particular, the amplified product will contain a target sequence in or complementary to a sequence in the HPIV genomic RNA, and a probe will bind directly or indirectly to a sequence contained in the amplified product to indicate the presence of HPIV nucleic acid in the tested sample.

(61) In embodiments that detect the amplified product near or at the end of the amplification step, a linear detection probe may be used to provide a signal to indicate hybridization of the probe to the amplified product. One example of such detection uses a luminescently labeled probe that hybridizes to target nucleic acid. Luminescent label is then hydrolyzed from non-hybridized probe. Detection is performed by chemiluminescence using a luminometer (see, e.g., International Patent Application Pub. No. WO 89/002476). In other embodiments that use real-time detection, the detection probe may be a hairpin probe such as, for example, a molecular beacon, molecular torch, or hybridization switch probe that is labeled with a reporter moiety that is detected when the probe binds to amplified product. Such probes may comprise target-hybridizing sequences and non-target-hybridizing sequences. Various forms of such probes are described, e.g., in U.S. Pat. Nos. 5,118,801; 5,312,728; 5,925,517; 6,150,097; 6,849,412; 6,835,542; 6,534,274; and 6,361,945; and US Patent Application Pub. Nos. 20060068417A1 and 20060194240A1).

EXAMPLES

(62) The following examples are provided to illustrate certain disclosed embodiments and are not to be construed as limiting the scope of this disclosure in any way.

(63) General Reagents and Methods. Unless otherwise indicated, amplifications were performed using an ABI 7500 FAST® instrument. Viral isolates used as amplification targets or controls were diluted in suitable media, e.g., Micro Test M4 media (Remel Inc. Cat. No. R12500), Micro Test M5 Viral Transport Medium (Remel, Inc. Cat. No. R12515), Micro Test M6 Viral Transport Medium (Remel, Inc. Cat. No. R12530), Micro Test M4RT Viral Transport Medium (Remel, Inc. Cat. No. R12505), or Copan Universal Transport Medium (Copan Diagnostics, Inc., Cat. No. 330C). Nucleic acid was extracted from viral isolates using a non-specific target capture procedure as described in US Patent App. Pub. 2013/0209992.

(64) Results for HPIV-1 through 3 were compared to results obtained in parallel with the Prodesse® ProParaflu+™ Assay, which is discussed in Loeffelholz et al., J. Clin. Microbiol. 49:4083-4088 (2011).

(65) Results for HPIV-4 (including subtypes 4a and 4b) were compared to an FDA cleared HPIV-4 assay (eSensor® Respiratory Virus Panel (GenMark Dx), xTAG™ Respiratory Viral Panel (Luminex), or Biofire FilmArray® Respiratory Panel (BiofireDiagnostics)). The FilmArray® Respiratory Panel is also discussed in Loeffelholz et al., supra. Samples previously characterized as positive for HPIV-4 by one of those three FDA cleared assays were tested with oligomers of the present disclosure. Samples that were discrepant were compared to results generated during development using an alternate buffer and enzyme system on the Cepheid SmartCycler II.

(66) PCR reactions were typically assembled as follows: 19.05 uL Supermix (Promega GoTaq® Supermix) 0.35 uL Roche MMLV Reverse Transcriptase (35 U) 0.6 uL GoTaq MDX Hotstart Taq (3U) 5 uL of nucleic acid (primers, probe, and target in suitable diluent) =25 uL total reaction volume.

Example 1—Oligomer Design Considerations

(67) For HPIV-1, HPIV-2, and HPIV-3, the hemagglutinin-neuraminidase (HN) gene was targeted. The nucleocapsid protein gene (NP) was targeted for HPIV-4a & 4b detection. It was determined that HPIV-4a and 4b can be detected with the same set of primers and probes.

(68) Due to the lack of sequence conservation among respiratory virus strains and to accommodate known mismatches, degenerate bases and Non-Watson Crick (NWC) bases were considered, e.g., in primers and TaqMan™ probes. NWC base pairing (deoxyguanosine/deoxyuridine or deoxyguanosine/deoxyadenosine) and/or degenerate nucleotides (in which a population of oligomers has individual molecules with one or the other of a pair of bases at degenerate positions) were incorporated in certain positions based on a review of sequence data collected by the inventors. Using Non-Watson Crick bases facilitates the use of a single detection oligonucleotide and thus simplifies manufacturing and quality control, making for a more reproducible process, but can decrease the melting temperature (T_m) of annealed oligomers and assay sensitivity for targets containing the non-Watson Crick base pair(s). In some oligomers, 5-methylcytosine bases were used to increase the stability of the duplex by raising the T_m by about 0.5°-1.3° C. for each 5-methylcytosine incorporated in an oligonucleotide (relative to the corresponding unmethylated oligomer). It was found that cytosine methylation could generally remedy the relatively lower stability of oligomers that used Non-Watson Crick base pairing.

Example 2—Evaluation of Non-Watson Crick Base Pairs

(69) For testing oligomers that use NWC base pairs, reference strains diluted suitable media were extracted as described above and in-vitro transcripts (IVTs) generated in house were diluted in RNase inhibitor/water. Each reference strain or IVT was tested at three anneal temperatures (57° C., 60° C. and 65° C.) with target-specific oligos containing either degenerate bases or Non-Watson Crick bases and 5-methylcytosine. The reference strains or IVTs were selected to match the various mutation combinations observed in the NCBI database at the location of the degenerate and Non-Watson Crick bases. For example, where an A and a G were observed in different strains, a U was used in the NWC oligomer (resulting in either a standard A-U pairing or a G-U wobble NWC pairing); and where a C and an A were observed in different strains, a G was used in the NWC oligomer (resulting in either a standard C-G pairing or a A-G purine-purine NWC pairing). NWC oligomers to be used as probes were designed to hybridize to the plus or minus strand of the target HPIV amplicon as appropriate so that the oligomer would contain a U, G, or A at the NWC position. NWC oligomers to be used as primers were designed to hybridize at sites where the variants or complements thereof predominantly contained either U and A, G and A, C and A, or C and U, or U and G in the strand to which the oligomer hybridizes.

(70) Testing of primers containing Non-Watson Crick base pairing demonstrated that 5-methylcytosine bases increased the T_m and binding stability of the primers to the DNA during PCR such that they produced Ct values similar to those found when using primers with degenerate bases. For example, implementing NWC base pairing and cytosine methylation gave an increase in the range of 0-1.9 Cts for HPIV-2 and 0-3.2 Cts for HPIV-3 compared to the degenerate base oligonucleotides at 57° C. and 60° C. anneal temperatures. At a 65° C. anneal temperature, the NWC oligomers produced Cts that were equal to or earlier than the degenerate base oligonucleotides. In many cases, there was no detection with the degenerate base oligonucleotides at 65° C.

(71) For the evaluation of Non-Watson Crick versus degenerate bases, HPIV-2 and HPIV-3 probes for the Non-Watson Crick oligonucleotides utilized license-free fluorophores and quenchers whereas degenerate base oligonucleotides utilized licensed fluorophores and quenchers obtained from Biosearch. This was due to a limited amount of license-free probes available. An increase in Ct values has been observed with the license-free fluorophores when compared to the licensed fluorophores and quenchers and possibly accounts for the Ct shift observed in the HPIV-2 and HPIV-3 oligonucleotides containing Non-Watson Crick bases. The HPIV-4 degenerate and NWC reactions utilized the same fluorophore/quencher

pair and the Ct shift was 1.4-3.1 Cts at 57° C. and only 0.4-1.4 Cts at 60° C.

(72) When license-free fluorophores and quenchers were available for both the Non-Watson Crick/5-methylcytosine probe and the degenerate base probe for HPIV-3, they were run with a 57° C. anneal step to test this possibility. Each HPIV-3 IVT or reference strain was tested in triplicate with both oligonucleotides. The average Ct value for the samples tested with the Non-Watson Crick/5-methylcytosine oligonucleotides were consistently earlier than the Cts for samples tested with the degenerate base oligonucleotides. One IVT was positive in one out of three replicates with the degenerate base oligonucleotides, but was positive in all three replicates with the Non-Watson Crick/5-methylcytosine oligonucleotides. All other samples were positive with both mixes. The total number of unique HPIV-3 samples tested was seven.

(73) The incorporation of the Non-Watson Crick bases with 5-methylcytosine is an acceptable solution that avoids using degenerate bases, which can complicate manufacturing and Quality Control testing relative to oligomers with only non-degenerate positions. It is also possible to use oligomers containing 5-methylcytosine in one, some (e.g., about half) or all positions, without degenerate bases and/or without bases designed to engage in NWC pairing, that can tolerate mismatches in one or a few (e.g., 2 or 3 or 4) bases (due to interstrain sequence variation among viruses) and still function as oligomers, e.g., primers and/or probes.

Example 3—Clinical Specimen Detection and Reactivity

(74) The clinical sensitivity of candidate oligomers was evaluated as described above in the General Reagents and Methods Section using exemplary primer pairs and probes of this disclosure. Seventy-three HPIV-1, 114 HPIV-2, 192 HPIV-3, and 54 HPIV-4 samples were extracted and tested. HPIV-4 samples were also tested with the Cepheid SmartCycler II and reagents obtained from Promega to compare the HPIV-4 clinical specimen detection results and resolve discrepancies.

(75) One HPIV-2 sample was negative upon initial testing with oligomers of this disclosure and 1 HPIV-2 sample was negative with the Prodesse® ProParaflu+™ Assay. Both samples were retested and were positive with the oligomers of this disclosure.

(76) Five HPIV-3 samples were detected by oligomers of this disclosure that were missed by the Prodesse® ProParaflu+™ Assay.

(77) The HPIV-4 samples were previously characterized by an FDA-cleared assay as noted above. During initial testing, 7 samples were negative with oligomers of this disclosure. The HPIV-4 detection was then modified to improve sensitivity and, with a lower threshold and an additional 50 mM KCl (total 100 mM), five out of seven HPIV-4 samples were positive. The hit rate for the HPIV-4 detection of oligomers of this disclosure was 52/54 or 96.3%. The two HPIV-4 negative samples had been previously characterized as HPIV-4 positive during earlier development using the same primers and probes when tested on the Cepheid SmartCycler II with an alternate buffer/enzyme system. The stability of the nucleic acid may have been compromised due to multiple freeze/thaws; therefore the samples were extracted again and tested with the oligomers of this disclosure in duplicate with additional KCl (100 mM total). These replicates were each positive for HPIV-4.

(78) TABLE-US-00002 TABLE 1 Hit rate analysis for HPIV clinical samples. PPF+: Prodesse® ProParaflu+™ Assay. IVD Cleared: FDA-cleared assay for HPIV-4 as described above. HPIV-1: Oligomers of this disclosure PPF+ # Detected: 73 73 Total: 73 73 Hit Rate: 100.00% 100.00% HPIV-2: Oligomers of this disclosure PPF+ # Detected: 114 113 Total: 114 114 Hit Rate: 100.00% 99.12% HPIV-3: Oligomers of this disclosure PPF+ # Detected: 192 187 Total: 192 192 Hit Rate: 100.00% 97.40% HPIV-4: Oligomers of this disclosure IVD Cleared # Detected: 54 54 Total: 54 54 Hit Rate: 100.00% 100.00%

(79) Due to a limited number of available Human Parainfluenza strains available, one HPIV-1, HPIV-2, HPIV-4a and HPIV-4b, one HPIV-4 (no subtype available) and two HPIV-3 cultured and titrated clinical isolates were extracted as described above to test reactivity. Each clinical isolate was tested using the oligomers of this disclosure and the Prodesse® ProParaflu+™ Assay (HPIV-1, HPIV-2 and HPIV-3) at two to three different concentrations in replicates of six on the ABI 7500 FAST®.

(80) The LoD for the additional strains between the oligomers of this disclosure and the Prodesse® ProParaflu+™ Assay was the same for HPIV-1 (10.sup.1 TCID.sub.50/ml) and HPIV-2 (10.sup.3 TCID.sub.50/ml). One of the HPIV-3 strains had the same LoD (10.sup.1 TCID.sub.50/ml) with both assays whereas the other was at least 2 logs better with the oligomers of this disclosure (10.sup.1 TCID.sub.50/ml) compared to the Prodesse® ProParaflu+™ Assay, which detected zero replicates at 10.sup.2 TCID.sub.50/ml, the highest concentration tested. LoDs were established for HPIV-4a, HPIV-4b and one HPIV-4 strain that was not designated as subtype a or b. Table 2 contains the reactivity data. Target nucleic acids were isolated from previously characterized stock organisms obtained from TriCore Reference Laboratories (Albuquerque, NM); ZeptoMetrix Corporation (Buffalo, NY); and ATCC (Manassas, VA).

(81) TABLE-US-00003 TABLE 2 Reactivity data. PPF+: Prodesse® ProParaflu+™ Assay. LoD & Average LoD & Average Ct Strain Ct PPF+ Oligomers of this disclosure HPIV-1 10.sup.1 TCID.sub.50/ml, 10.sup.1 TCID.sub.50/ml, Avg Ct = 39.4 Avg Ct = 37.1 HPIV-2 10.sup.3 TCID.sub.50/ml, 10.sup.3 TCID.sub.50/ml, Avg Ct = 37.4 Avg Ct = 36.8 HPIV-3 isolate 1 Negative at 10.sup.2 10.sup.1 TCID.sub.50/ml, Avg Ct = 38.9 HPIV-3 isolate 2 10.sup.1 TCID.sub.50/ml, 10.sup.1 TCID.sub.50/ml, Avg Ct = 39.0 Avg Ct = 36.0 HPIV-4a NA 10.sup.1 TCID.sub.50/ml, Avg Ct = 37.3 HPIV-4b NA 10.sup.2 TCID.sub.50/ml, Avg Ct = 34.6 HPIV-4 (subtype NA 10.sup.2 TCID.sub.50/ml, Avg Ct = 35.0 not specified)

(82) The oligomers used for HPIV-3 hybridized to positions encompassing nucleotides 1355, 1380, and 1437 (forward primer, probe, and reverse primer, respectively). The sites to which the oligomers hybridized for HPIV-3 were positions 1349-1371, 1372-1398, and 1426-1448 (forward primer, probe, and reverse primer, respectively); with regard to the numbering of SEQ ID NO: 3, the forward primer, probe, and reverse primer, respectively, for HPIV-3 hybridized to positions 249-271, 272-298, and 326-348. The sequences were SEQ ID NOs: 28, 33, and 37 (forward primer, probe, and reverse primer, respectively). SEQ ID NOs: 34-35 and 31-32 target the same locus as, and may perform similarly to, SEQ ID NO: 33.

Example 4—Analytical Sensitivity

(83) Reference strains were diluted in suitable media, spiked into pooled negative nasopharyngeal swab samples at 4 concentrations and extracted as described above. Each concentration was extracted in 6 replicates and the nucleic acids from the replicates were not combined. From each extraction, 3-4 replicates were used to test the oligomers of this disclosure and the Prodesse® ProParaflu+™ Assay for a total of 20 replicates tested with each assay. Assays were performed on the ABI 7500 FAST®.

(84) The Limit of Detection (LoD) for HPIV-1, HPIV-2 and HPIV-3 was approximately the same between the oligomers of this disclosure and the Prodesse® ProParaflu+™ Assay. The HPIV-1 LoD is 10.sup.1 TCID.sub.50/ml, HPIV-2 is 10.sup.0 TCID.sub.50/ml and HPIV-3 is 10.sup.1 TCID.sub.50/ml. The LoD of HPIV-4a and 4b was established as 10.sup.2 TCID.sub.50/ml and 10.sup.0 TCID.sub.50/ml, respectively.

(85) TABLE-US-00004 TABLE OF SEQUENCES In the following table, IUPAC nucleotide codes are used to identify degenerate (mixed) positions (Y = C or T, R = A or G, W = A or T, S = G or C, K = G or T, M = A or C, etc.) in which individual molecules in a composition or kit may have any of the nucleotides corresponding to the IUPAC code. SEQ ID

Descrip- NO tion Sequence 1 Exemplary
TGAAGACAAACACCATGACAGTATCCTCCGTGAACGAAAGTGCCAAAACAATCAAAGAGACAATCACAGAATTAA
HPIV-1

TCAGACAAGAAGTGATATCAAGGACTATAAACATACAAAGTTCAGTACAAAGCGGGATCCCAATATTGTTAAACA
sequence,
AGCAAAGCAGAGATCTCACACAATTAATAGAGAAGTCATGCAACAAACAGGAATTGGCTCAGATATGCGAGAACA
positions
CTATTGCTATTCCACCATGCAGACGGAATAACCCCTCTGGACCCACATGATTTCTGGAGATGTCCCGTAGGAGAAC
201-1200

CCCTATTGAGCAACAACCCTAATATCTCATTATTACCTGGACCAAGTCTACTTTCTGGATCTTCCACAATTTTCAG
GATGTGTCAGACTACCTTCATTATCAATTGGTGATGCAATATATGCGTATTCATCAAACCTTAATCACTCAAGGAT
GTGCAGATATAGGGAAGTCATATCAGGTTTTACAATTAGGTTACATATCTTTAAATTCAGATATGTATCCTGATT
TAAACCTGGTAATTTCTCATACCTATGACATCAATGACAACAGGAAATCATGTTCTGTAATAGCTGCAGGAACAA
GGGGTTACCAGTTATGCTCCTTGCCCACTGTGAATGAGACTACAGATTACTCGAGTGAAGGTATAGAAGACTTAG
TATTTGACATATTAGATCTCAAAGGAAAGACCAAATCTCATCGATACAAAAATGAAGATATAACTTTTGACCATC
CTTTTTCTGCAATGTATCCAAGTGTAGGAAGTGGGATAAAGATTGAAGATACACTCGTTTTCTTAGGATATGGTG
GCTTAACAACTCCGCTTCAAGGCAACACCAAGTGTGTGATAAGCAAATGTCCCAATGTTAATCAGAGTGTTCGA
ATGATGCTCTTAAGATAACTTGGCTAAAGAAAAGACAAGTTGTCAATGTCTTAATTCGTATCAATAATTATTTAT
CTGATAGGCCAAAGATTGTTGTGCGA 2 Exemplary

ATCATTGAGGCTCAATGGGTACCGTCCTATCAAGTTCCTCGTCCTGGAGTCATGCCATGCAATGCAACAAGTTTT
HPIV-2

TGCCCTGCTAATTGCATCACAGGGGTGTACGCAGATGTGTGGCCGCTTAATGATCCAGAACTCATGTACGTAAT
sequence,
GCTCTGAACCCCAACTATCGATTTGCTGGAGCCTTTCTCAAAAATGAGTCCAACCGAACTAATCCCACATTCTAC
positions
ACTGCATCGGCTAACTCCCTCTTAAATACTACCGGATTCAACAACACCAATCACAAAGCAGCATATACATCTTCA
1401-1900

ACCTGCTTTAAAAACACTGGAACCCAAAAAATTTATTGTTTAATAATAATTGAAATGGGCTCATCTCTTTTAGGG
GAGTTCCAAATAATACCATTTTTAAGGGAATAATGCTTTAATCCTATTGAATGAAGACTCCAGATTCAAGAATA
ATTGGAAGGCTCTTTATTTTATGCGATAGTTATACGTTTTGGCTGTATTA 3 Exemplary

TCATAGTCCATGGTTTTTCAGATAGAAGGATGGTCAATTCTATAATTGTTGTTGACAAGGGCTTAAACTCAGTTCC
HPIV-3

AAAATTGAAGGTATGGACGATATCTATGAGACAAAATTACTGGGGGTCAGAAGGAAGATTACTTCTACTAGGTAA
sequence,
CAAGATCTACATATACACAAGATCTACAAGTTGGCACAGCAAGTTACAATTAGGAATAATTGACATTACTGACTA
positions
CAGTGACATAAGGATAAAATGGACATGGCATAATGTGCTATCAAGACCAGGAAACAATGAATGTCCATGGGGACA
1101-1600

TTCATGTCCGGATGGGTGTATAACAGGAGTATATACTGATGCATATCCACTCAATCCCACAGGAAGCATTGTATC
ATCTGTCATATTGGACTCACAAAATCGAGAGTCAACCCAGTCATAACTTACTCAACAGCAACCGAAAGGGTGAA

CGAGTGGCCATTGCGAACAAAACACTCTCAGCTGGGTATACAACAACAA 4 Exemplary
TCCAGAGATTTGCCTATCAGCCTTGCCGATAGGACAGTATTTGTATCTAGAGAAGCAGAACATGCAGTGTGTGAC
HPIV-4
GATATGGATACATACCTGAATAGGATATTCAGTGTATTAATACAGATTTGGATAATGGTGTGCAAATGTATGACA
sequence,
GCCTATGATCAACCCACTGGATCTGAAGAGAGAAGGTTGGCTAAGTACAAGCAGCAGGGCAGGATGTTAGAGAAA
positions
TATCAACTGCAAACAGATGCAAGGAAAATTATCCAATTAGTAATAAGAGAGAGTATGGTTATAAGACAATTTCTT
501-12100
GTACAGGAGATGTTAACAGCTGATAAAGTAGGTGCTTATACTAATAGATATTATGCTATGGTAGGAGATATTGCA
AAGTATATAGCCAATGTCGGAATGAGTGCATTCTTTCTTACACTCAAGTTCGGTTTGGGTAAACAGGTGGAAACCA
CTTGCTCTAGCTGCATTTTCTGGTGAGTTAGTTAAACTAAAGTCTCTTATGTCATTATACAGAAGATTAGGTGAC
AGATCAAGATATTTGGCTCTCTTAGAATCTCCTGAATTGATGGAATTTGCTCCAGCCAATTACCCGCTATTATTC
AGTTACGCCATGGGAGTTGGAAGTGTCCAGGATCCATTGATCAGAACTATCAGTTTGGAAGGAACTTCTTAAAC
ACCAGCTATTTTCAGTATGGTGTGAGACTGCAATGAAACACCAGGGTACAGTTGACCCTAAATTGGCCTTAGAG
CTTGGAATAACTGATGAAGACAGAGTGGATATTATGCAGTCCGTTGAAAAGCACATATCAGGAAAAGCTGGTGAT
GATATATCACAGCCTGCAGGTGCATTTGCAATGTCTCTAAACAGATCAGCTTTCATTAACAACAACACTTCCCAA
GACTTCTCAGGGGCAAGACTCTCCAACCTACGAGCAAGGATGGTCCGGCACAAATCAAGATGAAACGAGAGATGCC
TATCCAGAACCTACAATGCATAGACTCCAAAACATTGAATCAACTGACTCAGATCACAATGAACTACTGATGCCG
GAGCTTGAAAGCGACAGCAACCCCTTCAATCGTCCAAGATTCACAGTCAGAGCACCCTCATCCCAGAAATCTCG
CACCAGAATCCAAATACCAGAATGAGCAGAGACATCAACACAAGGGACAACATCAGAGCTGACCGTCAAAATACC
AATGAGGACCGAGGCAGCAACATTCCAGATGACATTCTTGAGATCTAGACAATTGACACATCAACAACCCAGCC
ATAGCTAGACAATAATCATAAACCGGGAAAATCTCAAAGATACACAATCGCACACATACAATGAAAGCAATCACT
TCTGTTCCACATCACAACCCAGAAATCCTGCCAAACGCAACATCACGACCAATCAACTGATGATACACAGTAGAA
TACAAGAACACAACCTGCATTGATTTAAGAAAAAACTAAGATAGGCCTGAACGGTTGCATTGAGGATTCTCAATCG
TTCAGGCCAATACAATTATCAGCTTCAAACGAAAACAACAAGATGAGCTTTGAAATAAGCATGGAGGAGATTGAC
GAGCTCATTGAAACTGGCAATCTCAACATAGATTATGCACTTAAAGAATTAGGTGCAACCAGTCAACTCCCGCCA
AATAAATCTCCGTCTCAGACCAGCAAAACAGAAGAAACCAACGATGAGACAAGAACCTCAAAAAACTTGGCATCA
GGAGAGGCACCAGCCCACGCCTCTTCAACCTGCGGTCACACAATGAAGAGAGTGAATCAGGGAAACAGAGCCCCA
GACGGTTTCTCCATGATATCCAACAGACCCCAAACAGGTACACTGCTCATGGGATCAGACACACAATCTCCAAGT
CCATCAAAGACCTATCAAGGACTCATTCTTGATGCAAAGAAGAGAGCGCTAAATGAACCAAGGAGGAATCAAAAA
ACAACAAATGAACATGAAACACAAATGACACCAGGATATTTAAGAGGGGGGAATATAGCCACCAAGAAAGAGGC
TTGGGTACACAGGATCAGAGATCAAAAACGCAATCTTCATTCCAAGACATCGAAGGGAATACTCGATTTTCATGG
GTCAACGGAAGAACCACAATATCAGAGTGGTGCAATCCATGTTGCGCACCAAGTCAAACCAACCGCCTCTGTGCGAA
AAGTGTACATGTGGAAGATGTCCCAAAGTTTGCGAATTGTGCATCAGAGATCCTTGATGCAATCAAAGCATTAGA
GGTGAGGTTAGACAGGATCGAAGGGAAAGTTGACAAAATTATGCTTACTCAAATAACAATTCAGCAAAACAAAAA
TGACACTCAGCAAATTAAAGGCTCACTTGCAACAATTGAGGGCCTAATCACAACAATGAAAATAATGGACCCTGG
AGTCCCATCAAAAGTAAGTCTTAGAAGCTTAAACAAAGGGCCAGAGCAAGTTCCAATAATCGTCACTGGTACTGG
CGATGTCTCAAAATTTGTTGATCAAGACAATACAATTACACTTGATCCGTTAGCAAGACCCATTTTGTCTGGAAC
CAAGCAAACAACCTGATGAAAGGAGAGCAGGTGTTTCGTATAGATGCACTTAAATAACAGTATCAGAAATGATTCG
AGATCTATTTGGAGACTGTGATAAGAGCAGAAAACCTTCTTGAATCAATAAACATGGCAACCACAGAGCAAGAAAT
CAACTTGATCAAAACTAATGCTCTTAGAAGCATCACTTAAACTATGAGATCTATGAATCACGAAGGTCACAGCGA
CCCAGATTCACTCAATACGCCAAGCGCTCAACCCACACCCAGAAGGGACAACCGCACCGCATCCCCAAGAATCG
ACACCAGCTCAGACGACGGCAACACCAATCATCACAACAAAACACAGAACATCAGCACTGCATACAAAATCAC
CCCCCAGCACACACAATCCATCGACCGACAGATTCCCTCCATAACATTACAAAGTATTTAAGAAAAAAAACCAGAA
GTAAGGGGGGACCCATCATCCAATCGACTGTGGCCATGGCTCCCACTCAATCCAAAGTCAAAATTCACAATCTAG
CTGAAGCTCACGAGAAGGTACTGAGAGCCTTCCCAATAGAAGTTGAACAGAACTCAGAAGGGAAGAACTATTAG
TGAAACAGATCAGAATAAGAACGCTAGGACATGCGGATCATTCAAATGATTCAATTTGCTTTCTGAATACTTACG
GGTTTATCAAGGAAGCAGTCAGCCAGACAGAATTCATGAGAGCAGGACAGAGACTAGAAAGCAAAAACACTCTGA
CTGCTTGATGTTACCCCTTTGGGCCTGGTCTTAACATAGGGTCCCCCCTAAAGATGCTAGAAATATGCAGAAGACA
TTAAGATCAATGTAAGAAAGACAGCAGGCTGCAAGGAACAGATTGTGTTTTCTCTAGATAGGACACCTCAAGTGT
TCAGGGGATTTCAATTTCCCCGGGATAGATATGCATGTGTTCCATCTGACAAATATATAAAATCACCAGGAAAGT
TAGTCGCAGGACCCAATTATTGTTACATTATCACTTTCCTTTTCATTGACATTCTGTCCCAGCAGCCAGAAATTCA
AAGTACCTAGACCCATATTGAATTTTCGATCCACACGAATGAGGGGTATACATCTAGAGATTATCATGAAAATTA
CGTGTCTTGAAAACAGTCCAATCAGAAAGACACTAATAACAGATGATCCCGAAAATGGACCAAAAGCATCCGTGT
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 CAGAGATCTUACACAATTAATAGAGAAGTCATGCAAC sense oligomer

Claims

1. A composition comprising at least a first HPIV-2 amplification oligomer and a second HPIV-2 amplification oligomer, wherein: the first HPIV-2 amplification oligomer and second HPIV-2 amplification oligomer are configured to amplify an HPIV-2 target nucleic acid sequence of at least about 50 nucleotides in length comprising at least one HPIV-2 position located within HPIV-2 positions 1600-1700, wherein the first HPIV-2 amplification oligomer comprises the sequence of SEQ ID NO: 44, and wherein the composition further comprises a third HPIV-2 oligomer configured to hybridize to the HPIV-2 target nucleic acid sequence, and the third oligomer comprises a chemiluminescent label or a fluorescent label.
2. The composition or kit of claim 1, wherein the second HPIV-2 amplification oligomer is configured to hybridize to a site comprising HPIV-2 position 1698.
3. The composition of claim 2, wherein the second HPIV-2 amplification oligomer competes for hybridization to an HPIV-2 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 40.
4. The composition of claim 3, wherein the second HPIV-2 oligomer comprises the sequence of SEQ ID NO: 40 or a sequence with one or two mismatches to SEQ ID NO: 40.
5. The composition of claim 4, wherein the second HPIV-2 oligomer comprises the sequence of SEQ ID NO: 40.
6. The composition of claim 1, wherein the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 39, 41, 42, 43, or 45 or a sequence with one or two mismatches to SEQ ID NO: 39, 41, 42, 43, or 45.
7. The composition of claim 6, wherein the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 39.
8. The composition of claim 7, wherein the third HPIV-2 oligomer comprises the sequence of SEQ ID NO: 38.
9. The composition or kit of claim 1, wherein at least one of the first, second, and third HPIV-2 oligomers comprises at least one 5-methylcytosine.
10. A method of detecting HPIV-2 in a sample, comprising: contacting the sample with at least a first HPIV-2 amplification oligomer and a second HPIV-2 amplification oligomer, wherein the first HPIV-2 amplification oligomer and second HPIV-2 amplification oligomer are configured to amplify an HPIV-2 target nucleic acid sequence of at least about 50 nucleotides in length comprising at least one HPIV-2 position located within HPIV-2 positions 1600-1700, wherein the first HPIV-2 amplification oligomer comprises the sequence of SEQ ID NO: 44, performing a nucleic acid amplification reaction which produces an HPIV-2 amplicon in the presence of an HPIV-2 nucleic acid, and detecting the presence or absence of the HPIV-2 amplicon.
11. The method of claim 10, wherein the second HPIV-2 amplification oligomer is configured to hybridize to a site comprising HPIV-2 position 1698.
12. The method of claim 11, wherein the second HPIV-2 amplification oligomer competes for hybridization to an HPIV-2 nucleic acid under stringent conditions with an oligomer having a sequence consisting of SEQ ID NO: 40.
13. The method of claim 12, wherein the second HPIV-2 oligomer comprises the sequence of SEQ ID NO: 40 or a sequence with one or two mismatches to SEQ ID NO: 40.
14. The method of claim 13, wherein the second HPIV-2 oligomer comprises the sequence of SEQ ID NO: 40.
15. The method of claim 10, wherein the nucleic acid amplification reaction comprises PCR with a polymerase with 5'-to-3' exonuclease activity, and detecting the HPIV-2 amplicon using a probe oligomer comprising a fluorophore and a quencher, wherein exonucleolysis of the probe by the polymerase reduces quenching of fluorescence by the quencher.
16. A reaction mixture for detecting HPIV-2 in a sample, comprising: (i) at least a first HPIV-2 amplification oligomer and a second HPIV-2 amplification oligomer, wherein the first HPIV-2 amplification oligomer and second HPIV-2 amplification oligomer are configured to amplify an HPIV-2 target nucleic acid sequence of at least about 50 nucleotides in length comprising at least one HPIV-2 position located within HPIV-2 positions 1600-1700, wherein the first HPIV-2 amplification oligomer comprises the sequence of SEQ ID NO: 44 and/or the second HPIV-2 amplification oligomer comprise the sequence of SEQ ID NO: 40, and (ii) a third HPIV-2 oligomer configured to hybridize to the HPIV-2 target nucleic acid sequence, wherein the third oligomer comprises a chemiluminescent label or a fluorescent label, optionally wherein the reaction mixture further comprises at least one enzyme selected from the group consisting of a DNA polymerase, a reverse transcriptase, and an RNA polymerase, and wherein the reaction mixture is aqueous or lyophilized.

17. A composition comprising at least a first HPIV-2 amplification oligomer and a second HPIV-2 amplification oligomer, wherein: the first HPIV-2 amplification oligomer and second HPIV-2 amplification oligomer are configured to amplify an HPIV-2 target nucleic acid sequence of at least about 50 nucleotides in length comprising at least one HPIV-2 position located within HPIV-2 positions 1600-1700, wherein the second HPIV-2 amplification oligomer comprises the sequence of SEQ ID NO: 40, and wherein the composition further comprises a third HPIV-2 oligomer configured to hybridize to the HPIV-2 target nucleic acid sequence, and the third oligomer comprises a chemiluminescent label or a fluorescent label.

18. The composition of claim 17, wherein the first HPIV-2 amplification oligomer is configured to hybridize to a site comprising HPIV-2 position 1603 and competes for hybridization to an HPIV-2 nucleic acid under stringent conditions with an oligomer having the sequence consisting of SEQ ID NO: 44.

19. The composition of claim 18, wherein the first HPIV-2 amplification oligomer comprises the sequence of SEQ ID NO: 44 or a sequence with one or two mismatches to SEQ ID NO: 44.

20. A method of detecting HPIV-2 in a sample, comprising: contacting the sample with at least a first HPIV-2 amplification oligomer and a second HPIV-2 amplification oligomer, wherein the first HPIV-2 amplification oligomer and second HPIV-2 amplification oligomer are configured to amplify an HPIV-2 target nucleic acid sequence of at least about 50 nucleotides in length comprising at least one HPIV-2 position located within HPIV-2 positions 1600-1700, wherein the second HPIV-2 amplification oligomer comprises the sequence of SEQ ID NO: 40, performing a nucleic acid amplification reaction which produces an HPIV-2 amplicon in the presence of an HPIV-2 nucleic acid, and detecting the presence or absence of the HPIV-2 amplicon.

21. The method of claim 20, wherein the first HPIV-2 amplification oligomer is configured to hybridize to a site comprising HPIV-2 position 1603 and competes for hybridization to an HPIV-2 nucleic acid under stringent conditions with an oligomer having the sequence consisting of SEQ ID NO: 44.

22. The method of claim 21, wherein the first HPIV-2 amplification oligomer comprises the sequence of SEQ ID NO: 44 or a sequence with one or two mismatches to SEQ ID NO: 44.
